

FieldsI-Notes

Quantum Field Theory I

Spring 2019

January 2019 - March 2019

1 Lecture 1 (January 29)

The subject of Field Theory is the dynamical systems (classical or quantum) with continuously many degrees of freedom. Typically, there is some number of degrees of freedom associated with each point of space $\mathbf{x} \in \mathbf{R}^3$. Examples are:

i) Electromagnetic Theory

Degrees of Freedom : $\mathbf{E}(\mathbf{x}, t), \mathbf{H}(\mathbf{x}, t)$.

ii) Gravitation (general Relativity)

Degrees of Freedom = Spacetime metric $g_{\mu\nu}(\mathbf{x}, t)$.

iii) More exotic example: Dirac (electron-positron) field

Degrees of Freedom = Four – component (complex) Lorentz spinor $\psi_a(\mathbf{x}, t)$.

Main subject of this course is Quantum Theory of Fields. The states of quantum fields with local interaction admit interpretation in terms of propagating and interacting particles. Thus the theory of quantum fields is the basis of particle theory.

Classical Field Theory

We start with a brief review of basic aspects of classical field theory. Fundamental object in classical mechanics is the **action**,

$$S = \int L(q(t), \dot{q}(t)) dt;$$

the equations of motion are derived from the extremal action principle $\delta S = 0$. In field theory the coordinates are replaced by the field degrees of freedom,

$$q(t) \rightarrow \Phi(\mathbf{x}, t),$$

and the action is written as a space-time integral

$$S = \int \mathcal{L}(\Phi(\mathbf{x}, t), \dot{\Phi}(\mathbf{x}, t), \nabla\Phi(\mathbf{x}, t)) d^3\mathbf{x} dt.$$

Here $\Phi = \Phi(\mathbf{x}, t)$ stands for a collection of fields (functions of $\mathbf{x}, t$, representing local degrees of freedom), which may include scalar fields, components of vectors, tensors, spinors, etc. It is conventionally assumed that Φ has finitely many individual components. The function $\mathcal{L}$ is called the **Lagrangian density**, it depends on $\Phi(\mathbf{x}, t)$ and its derivatives over $\mathbf{x}$ and t , taken at the same space-time point $(\mathbf{x}, t)$. Two remarks are in order.

1. The above expression is suitable for imposing conditions of the special relativity. The integration is over the 4-dimensional Minkowskian space-time,

$$d^3\mathbf{x} dt = d^4x .$$

Recall that in relativistic theory one deals with the 4-dimensional space-time with the coordinates

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, \mathbf{x}) ,$$

equipped with the “pseudo-metric”

$$d\tau^2 = c^2 dt^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 .$$

The Lorentz transformations are those linear transformations of the coordinates x^μ which preserve $d\tau^2$. In what follows I will use the units in which¹

$$c = 1 , \quad \hbar = 1 .$$

Then

$$d\tau^2 = dx^\mu dx_\mu , \quad \text{where} \quad x^\mu = (t, \mathbf{x}) , \quad \text{and} \quad x_\mu = (t, -\mathbf{x}) .$$

To ensure the relativistic invariance of the classical field theory it suffices to choose $\mathcal{L}$ to be a Lorentz scalar.

2. In writing the above action I already accepted severe restrictions, namely,

$$S = \int dt L , \quad \text{with}$$

¹These conditions are simply the convention to measure spatial distances and energy in the units

$$[\text{unit of length}] = c \times [\text{unit of time}] , \quad [\text{unit of energy}] = \hbar / [\text{unit of time}] .$$

$$L = \int d^3\mathbf{x} \mathcal{L}(\Phi(\mathbf{x}, t), \dot{\Phi}(\mathbf{x}, t), \nabla\Phi(\mathbf{x}, t)) ,$$

i.e. L is a $d^3\mathbf{x}$ integral of a *spatial density* $\mathcal{L}$, which depends on Φ , $\dot{\Phi}$, and $\nabla\Phi$, all taken *at the same point* $\mathbf{x}$. This means that the field degrees of freedom $\Phi(\mathbf{x})$ at some point $\mathbf{x}$ do not have a direct interaction with $\Phi(\mathbf{x}')$ located at some other point $\mathbf{x}'$, with $|\mathbf{x} - \mathbf{x}'|$ greater than zero. Allowing such interactions would lead to possible terms in L like

$$L_{\text{nonlocal}} = \int F(\Phi(\mathbf{x}, t), \Phi(\mathbf{x}', t)) d^3\mathbf{x} d^3\mathbf{x}'$$

(“nonlocal interactions”). Nonlocal interactions do not go well along with the relativistic invariance. For if S includes, say, double integrals over $\mathbf{x}$ (as above), relativistic invariance of S forces one to include the terms which are nonlocal in time as well, like

$$\int F'(\Phi(t), \Phi(t')) dt dt' ,$$

which evidently violate *causality*: The state in the future affects the dynamics now. Locality and causality are deeply connected.

The above local form, $L = \int \mathcal{L}(\Phi(\mathbf{x}, t), \dot{\Phi}(\mathbf{x}, t), \nabla\Phi(\mathbf{x}, t)) d^3\mathbf{x}$, describes the dynamics in which the interaction occurs only between the degrees of freedom associated with “infinitely close” spatial points (through the $\nabla\Phi$ in L).

Also, I did not include higher spatial derivatives of Φ as the arguments in $\mathcal{L}$. Although such dependence would not violate locality (on a superficial level), including such dependencies in a relativistic theory would require to add higher time derivatives, like $\ddot{\Phi}$, as well. Such dependence is not allowed in conventional Lagrangian dynamics, which requires that the time evolution $\varphi(\mathbf{x}, t)$ is completely determined by the initial values of $\Phi(\mathbf{x})$ and $\dot{\Phi}(\mathbf{x})$ at $t = t_0$.

Anyhow, we start with the above local form of the action. In classical field theory the field equations of motion follow from the action principle

$$\delta S = 0 .$$

Let us combine the spatial and time variable into the 4-coordinate $x = (x^\mu) = (x^0, \mathbf{x})$, with $x^0 = t$, and write $\Phi(x)$ instead of $\Phi(\mathbf{x}, t)$; the Lagrangian density then is written as

$$\mathcal{L}(\Phi(x), \partial_\mu\Phi(x)) ,$$

where

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \nabla \right).$$

A simple exercise in functional analysis (look up the textbook, page 15 in PS) yields the classical field equations of motion

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi(x))} \right) - \frac{\partial \mathcal{L}}{\partial \Phi(x)} = 0. \quad (1.1)$$

Remark: Essential step in deriving (1.1) is integrating by parts. The total derivative term

$$\int d^4x \frac{\partial}{\partial x^\mu} \left(\delta \Phi(x) \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi(x))} \right)$$

can be reduced to the integral over the boundary (Gauss theorem), which depends only on the variations $\delta \Phi(x)$ at the boundary; this term does not affect the "bulk" equations of motion (1.1), which hold at each point x of the space-time. But if the field dynamics is considered in a space-time domain D , with the boundary ∂D , the above term contributes to the variation of S via the variations of the boundary values $\delta \Phi(x) |_{x \in \partial D}$ of the fields,

$$\int_D d^4x \partial_\mu (...) = \int_{\partial D} \delta \Phi(x) \frac{\partial \mathcal{L}}{\partial \partial_\mu \Phi(x)} d\Sigma_\mu(x)$$

where $d\Sigma_\mu(x)$ is the normal² element of the boundary hypersurface ∂D . In this situation the total derivative term contributes to the boundary conditions (see the homework Problem 1). The default setup is $D =$ full infinite space-time, with the condition that $\Phi(x)$ decays sufficiently fast at both space and time infinities ("scattering asymptotic conditions"); in this setup the above derivative term can be ignored.

The task of the classical field theory is to find solutions of these equations with such and such initial or boundary conditions.

One of the most important general results about classical field theory is the relation between symmetries and conservation laws known as the

²"Normal" means that $d\Sigma_\mu(x) T^\mu(x) = 0$ for any $T^\mu(x)$ tangent to the boundary ∂D , at any point $x \in \partial D$.

Noether's theorem (pp 17-18 of PS)

What is a symmetry? Suppose we have a family of continuous transformations of the field variables

$$\Phi(x) \rightarrow \Phi'(x) = F(x, \Phi(x))$$

such that the action S does not change

$$S[\Phi] = S[\Phi'] .$$

The term continuous transformations means here that they depend on some continuous parameter(s), $F = F_s$, in such a way that

$$F_0(x, \Phi(x)) = \Phi(x) \quad - \text{no transformation} .$$

(discrete symmetries is a separate story). Then we can take an infinitesimal transformation

$$\Phi(x) \rightarrow \Phi'(x) = \Phi(x) + \epsilon E(x, \Phi(x)) ,$$

where

$$E = \left. \frac{d}{ds} F \right|_{s=0} ,$$

with an infinitesimal ϵ . Substituting in the Lagrangian density, one finds

$$\mathcal{L}(\Phi', \partial_\mu \Phi') = \mathcal{L}(\Phi, \partial_\mu \Phi) + \epsilon \left[\frac{\partial \mathcal{L}}{\partial \Phi} E(x, \Phi) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \partial_\mu E(x, \Phi) \right] + O(\epsilon^2) .$$

The invariance (i.e. the requirement that the action $S = \int \mathcal{L} d^4x$ does not change) implies

$$[\dots] = \partial_\mu \tilde{J}^\mu(\Phi, \partial_\mu \Phi, x) ,$$

with some $\tilde{J}^\mu$ which depend on our field variables in a local way (it may also have separate dependence on x if E does). Transforming the left-hand side by parts one finds

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} E(x, \Phi) \right) + \left[\frac{\partial \mathcal{L}}{\partial \Phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \right) \right] E(x, \Phi) = \partial_\mu \tilde{J}^\mu .$$

Now, suppose that $\Phi = \Phi_c$, a solution of the classical field equations of motion. Then the second term vanishes, and we obtain the continuity equation

$$\partial_\mu J^\mu = 0 ,$$

where

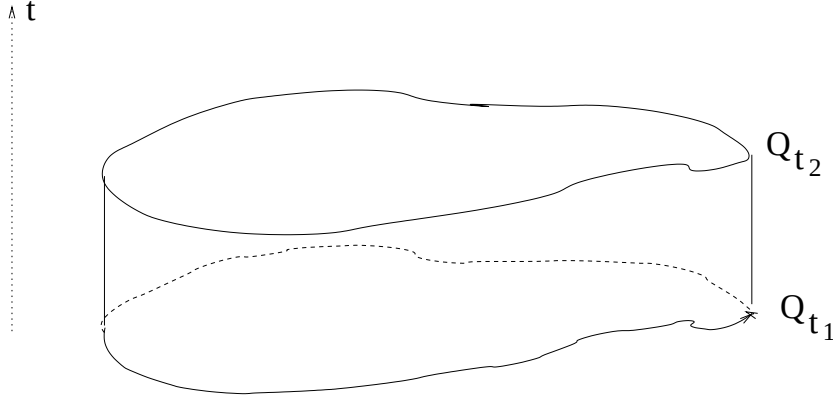
$$J^\mu(\Phi, \partial\Phi, x) = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)} E(x, \Phi) - \tilde{J}^\mu(\Phi, \partial\Phi, x) ,$$

which holds as long as Φ solves the classical field equations.

The continuity equation then implies a conservation law. Indeed, by the Gauss theorem

$$\oint_{\Sigma} J^\mu d\Sigma_\mu = \int_D \partial_\mu J^\mu d^4x = 0 .$$

Here $\Sigma = \partial D$ is any closed 3-dimensional hypersurface in the 4-dimensional space-time, and $d\Sigma_\mu$ is the normal element of $e_{\mu\nu\lambda\rho} dx^\nu dx^\lambda dx^\rho$ of this hypersurface. Taking the hypersurface to be a big slab between two equal-time hyperplanes and bringing the spatial boundary to the spatial infinity,



and also assuming that all fields decay at $\mathbf{x} \rightarrow \infty$, one finds

$$Q_{t_1} = Q_{t_2} ,$$

where

$$Q_t = \int J^0 d^3\mathbf{x} ,$$

that is Q is an integral of motion for the classical field equations. Let me stress again that both the continuity equation $\partial_\mu J^\mu = 0$ and the conservation law $\frac{d}{dt}Q = 0$ are satisfied only if we take $\Phi = \Phi_c$ (in terminology of QFT they are said to hold “on shell”).

Energy-Momentum

The most common symmetry we encounter in field theory is the translational symmetry. Assume that the Lagrangian density $\mathcal{L}(\Phi, \partial\Phi)$ has no x dependence except for that coming through the field variable $\varphi(x)$. It means that the dynamics looks the same if we shift the space-time coordinates

$$x \rightarrow x' = x + a$$

by a constant 4-vector a . This can be viewed as the symmetry with respect to the following transformation of the field variables

$$\Phi(x) \rightarrow \Phi'(x) = \Phi(x + a),$$

which leaves the action invariant. This field transformation has the infinitesimal form

$$\Phi'(x) = \Phi(x) + da^\mu \partial_\mu \Phi(x),$$

i.e. in this example $E(\Phi, \partial\Phi) = \partial_\mu \Phi$. When we do this transformation the action is going to stay invariant, but the Lagrangian density changes by the total derivative

$$\mathcal{L}' = \mathcal{L} + da^\nu \partial_\nu \mathcal{L},$$

where the derivative in the last term can be written as $\partial_\mu (\delta^\mu_\nu \mathcal{L})$, and has the same meaning as $\partial_\mu \tilde{J}^\mu$ in the general argument above.

Note that there are four independent symmetries associated with four components of a^μ . Correspondingly, there are four Noether's currents

$$T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \partial_\nu \Phi - \delta^\mu_\nu \mathcal{L},$$

which therefore satisfies the continuity equations

$$\partial_\mu T^\mu_\nu = 0 \quad \text{on shell.}$$

This object is known as the **energy-momentum tensor**. It leads to four conserved charges

$$E = \int T^0_0 d^3 \mathbf{x},$$

and

$$P^i = - \int T^0_i d^3 \mathbf{x}, \quad \text{with } i = 1, 2, 3.$$

Of these, E is interpreted as the energy of the field configuration $\Phi(\mathbf{x}, t)$, and $\mathbf{P}$ is interpreted as its momentum. Therefore one may take

$$T^{00} = T_0^0 = \mathcal{E} \quad \text{for the energy density,}$$

and

$$T^{0i} = -T_i^0 = \mathcal{P}^i \quad \text{for the momentum density.}$$

The spatial components T^{ij} constitute the (spatial) *stress tensor*. It is interpreted as usual: Consider any domain $D = D^3$ in the 3-dimensional space (at a given time). The momentum of the field inside this domain is

$$P_D^i = \int_D \mathcal{P}^i(\mathbf{x}) d^3\mathbf{x}.$$

According to the pulse-momentum theorem,

$$\frac{d}{dt} P_D^i = F_D^i, \quad \text{the net force applied to the field inside } D.$$

Since the interaction is local, this force must come from the surface forces,

$$F_D^i = - \int_{\partial D} T^{ij}(\mathbf{x}) d\sigma_j = - \int_D \partial_j T^{ij}(\mathbf{x}) d^3\mathbf{x}.$$

Thus, $dF^i = T^{ij}(\mathbf{x})d\sigma_j$ is the force exerted by the field on the surface element $d\sigma_j$.

Klein-Gordon Field

Let us consider a simple field theory known as the *Klein-Gordon* theory (it will be our main example for some time). It involves a single-component scalar field $\Phi(x) = \varphi(\mathbf{x}, t)$, and the action is

$$\begin{aligned} S &= \int d^4x \left[\frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{m^2}{2} \varphi^2 \right] \equiv \\ &\int dt d^3\mathbf{x} \left[\frac{1}{2} \dot{\varphi}^2 - \frac{1}{2} (\nabla \varphi)^2 - \frac{m^2}{2} \varphi^2 \right]. \end{aligned}$$

This form of the action shows that the theory is Lorentz-invariant, and the field φ is a scalar (i.e. $\varphi(x)$ has definite value at each space-time point x , independent of the choice of the inertial frame). Variation yields the Klein-Gordon field equation

$$\partial_\mu \partial^\mu \varphi + m^2 \varphi = \partial_t^2 \varphi - \nabla^2 \varphi + m^2 \varphi = 0.$$

Applying the above equations one finds for the energy-momentum tensor

$$T_{\mu\nu} = \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} (\partial_\lambda \varphi \partial^\lambda \varphi - m^2 \varphi^2),$$

where

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

is the usual Minkowski metric. Thus,

$$\mathcal{E} = \frac{1}{2} \left(\dot{\varphi}^2 + (\nabla \varphi)^2 + m^2 \varphi^2 \right),$$

and

$$\mathcal{P} = \dot{\varphi} \nabla \varphi.$$

We see in this example that the energy-momentum tensor $T_{\mu\nu}$ is symmetric,

$$T_{\mu\nu} = T_{\nu\mu}.$$

Exercise

Show that in any Lorentz-invariant theory involving only a scalar field $\varphi(x)$ the energy-momentum tensor always comes out symmetric, $T_{\mu\nu} = T_{\nu\mu}$. Check that in such theory all components of the antisymmetric tensor

$$M^{\mu\nu} = \int [x^\mu T^{\nu 0} - x^\nu T^{\mu 0}] d^3 \mathbf{x}$$

are conserved on-shell.

The equations of motion of the Klein-Gordon theory

$$\partial_\mu \partial^\mu \varphi + m^2 \varphi = \partial_t^2 \varphi - \nabla^2 \varphi + m^2 \varphi = 0.$$

are *linear*. That is, any linear combination of solutions are again the solutions. (In general, this property defines the *free field theory*). Therefore, general solution is a superposition of "elementary solutions". In the infinite space-time, one can take the plane waves

$$e^{-ipx} = e^{-ip^0 x^0 + i\mathbf{p}\mathbf{x}} \quad \text{with} \quad p^0 = \omega_{\mathbf{p}} := \sqrt{\mathbf{p}^2 + m^2}$$

as well as the complex-conjugate waves³

$$\varphi(x) = \int \frac{d^3\mathbf{p}}{2\omega_{\mathbf{p}}(2\pi)^3} (A_{\mathbf{p}} e^{-ipx} + A_{\mathbf{p}}^* e^{ipx}) . \quad (1.2)$$

This form represents general solution of the Klein-Gordon equation, which is bounded at the spatial infinity.

³As usual

2 Lecture 2 (January 31)

Before going to quantum theory, let us recall the Hamiltonian formalism of classical mechanics and see how it looks for fields.

Hamiltonian Formalism

In Lagrangian dynamics with the generalized coordinates q^i one passes to the Hamiltonian formalism through introducing the conjugate momenta

$$p_i = \frac{\partial L(q, \dot{q})}{\dot{q}^i},$$

and then excluding $\dot{q}$ in favor of p , by solving these equations. The Hamiltonian function H appears as the Legendre transform

$$H(p, q) = \sum_i p_i \dot{q}^i - L.$$

The equations of motion then take the canonical form

$$\dot{p}_i = \{H, p_i\} = -\frac{\partial H}{\partial q_i},$$

$$\dot{q}^i = \{H, q^i\} = \frac{\partial H}{\partial p_i},$$

in terms of the Poisson brackets

$$\{f, g\} := \sum_i \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} - \frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i}.$$

In a field theory we have continuously many generalized coordinates $\varphi(\mathbf{x})$ (I assume here a scalar field, for simplicity), labeled by the spatial coordinates $\mathbf{x}$. In this case the conjugated momenta are defined as

$$\pi(\mathbf{x}) = \frac{\delta L}{\delta \dot{\varphi}(\mathbf{x})},$$

with the basic Poisson brackets

$$\{\pi(\mathbf{x}), \varphi(\mathbf{x}')\} = \delta(\mathbf{x} - \mathbf{x}').$$

In the Klein-Gordon theory

$$\pi(\mathbf{x}) = \dot{\varphi}(\mathbf{x}) ,$$

and the Hamiltonian is

$$H = \int d^3\mathbf{x} \left[\frac{1}{2} \pi^2(\mathbf{x}) + \frac{1}{2} (\nabla\varphi)^2(\mathbf{x}) + \frac{m^2}{2} \varphi^2(\mathbf{x}) \right] . \quad (2.1)$$

Of course, it coincides with the above expression for the energy, $E = \int T^{00} d^3\mathbf{x}$, expressed through π rather than $\dot{\varphi}$.

Quantum Klein-Gordon field

We can try now to develop quantum theory of the Klein-Gordon field, following the usual rules of canonical quantization. One starts with the correspondence

Classical Theory	Quantum Theory
Phase space	Hilbert space $\mathcal{H}$
p_i, q^i	Hermitian operators $\hat{p}_i, \hat{q}^i : \mathcal{H} \rightarrow \mathcal{H}$
$H(p, q)$	$\hat{H}(\hat{p}, \hat{q}) : \mathcal{H} \rightarrow \mathcal{H}$

where the Hermitian operators $\hat{q}, \hat{p}$ must obey the canonical commutation relations

$$[\hat{q}^i, \hat{p}_j] = i \delta_j^i .$$

Remark: In QM with finitely many degrees of freedom, in the Schroedinger's quantization one can take the space of square-integrable functions $\Psi(q)$ for $\mathcal{H}$, and define the corresponding representation

$$\begin{aligned} \hat{q}^i \Psi(q) &= q^i \Psi(q) , \\ \hat{p}_i \Psi(q) &= -i \frac{\partial}{\partial q^i} \Psi(q) , \end{aligned}$$

and $\hat{H} = H(\hat{p}, \hat{q})$. The time evolution of the state $\Psi(q)$ is governed by the Schroedinger equation

$$i \frac{\partial}{\partial t} \Psi(q) = \hat{H} \Psi(q) .$$

In principle, one can follow this prescription and introduce the operators

$$\hat{\varphi}(\mathbf{x}) , \quad \hat{\pi}(\mathbf{x})$$

and demand that they satisfy the canonical commutators in the form

$$[\hat{\varphi}(\mathbf{x}), \hat{\pi}(\mathbf{x}')] = i \delta(\mathbf{x} - \mathbf{x}') .$$

Furthermore, one can take the space of functionals $\Psi[\varphi(\mathbf{x})]$ as the Hilbert space $\mathcal{H}$; it provides a formal representation of the above canonical commutation relations through the identifications

$$\begin{aligned}\hat{\varphi}(\mathbf{x}) \Psi[\varphi] &= \varphi(\mathbf{x}) \Psi[\varphi], \\ \hat{\pi}(\mathbf{x}) \Psi[\varphi] &= -i \frac{\delta}{\delta \varphi(\mathbf{x})} \Psi[\varphi].\end{aligned}$$

The Hamiltonian then is

$$\hat{H} = \int d^3\mathbf{x} \left[-\frac{1}{2} \frac{\delta^2}{\delta \varphi(\mathbf{x})^2} + \frac{1}{2} (\nabla \varphi)^2 + \frac{m^2}{2} \varphi^2 \right].$$

and it remains to find all solutions of the stationary Schroedinger equation

$$\hat{H} \Psi[\varphi] = E \Psi[\varphi].$$

Exercise: Show that this Schroedinger equation admits formal solution of the form

$$\Psi_0[\varphi] = \text{Const} \exp \left\{ \int \varphi(\mathbf{x}) G(\mathbf{x} - \mathbf{x}') \varphi(\mathbf{x}') d^3\mathbf{x} d^3\mathbf{x}' \right\}$$

Find the kernel $G(\mathbf{x} - \mathbf{x}')$. (At this point, do not worry when you discover divergent integrals). Try to look for further solutions in the form

$$\Psi_n[\varphi] \sim \int F(\mathbf{x}_1, \dots, \mathbf{x}_n) \varphi(\mathbf{x}_1) \dots \varphi(\mathbf{x}_n) \exp \left\{ \int \varphi(\mathbf{x}) G(\mathbf{x} - \mathbf{x}') \varphi(\mathbf{x}') d^3\mathbf{x} d^3\mathbf{x}' \right\}.$$

Although this straightforward approach, with certain refinements, would work in the Klein-Gordon theory, it has a number of unpleasant features. One is that the scalar product in $\mathcal{H}$ has to be defined through the functional integral

$$(\Psi_1, \Psi_2) \sim \int \Psi_1^*[\varphi] \Psi_2[\varphi] D[\varphi],$$

and certain care must be taken to give it a satisfactory definition (We are going to use functional integrals, in somewhat different context, though). This and other features of the above representation in terms of the variational derivatives makes this approach somewhat cumbersome. Much shorter route is based on explicit separation of variables, as we discuss below.

One observes that the Hamiltonian functional $H[\varphi]$ of the Klein-Gordon theory is quadratic in the field variables $\varphi(\mathbf{x})$ and their conjugate momenta $\pi(\mathbf{x})$; this means that the theory can be understood as the *collection of harmonic oscillators* (this observation actually suggests the Gaussian form of $\Psi_0[\varphi]$ in the above Exercise). It is then straightforward to make linear transformation of the canonical variables $\varphi(\mathbf{x}), \pi(\mathbf{x})$ which separates the variables into a system of non-interacting harmonic

oscillators, and then quantize the individual oscillators. This transformations are described in details in PS, and I refer to **Sect 2.3** there for the detailed calculation.

Instead, here I will follow equivalent routine based on special symmetries of the KG theory. Recall that the Lagrangian density is

$$\mathcal{L} = \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{m^2}{2} \varphi^2 ,$$

and observe that the transformation of the field

$$\varphi(x) \rightarrow \varphi'(x) = \varphi(x) + f(x) ,$$

where $f(x)$ is an arbitrary solution of the KG equation

$$\partial_\mu \partial^\mu f(x) + m^2 f(x) = 0$$

changes the Lagrangian density changes by a total derivative term

$$\mathcal{L}(\varphi', \partial\varphi') = \mathcal{L}(\varphi, \partial\varphi) + \partial_\mu \tilde{J}_f^\mu ,$$

where

$$\tilde{J}_f^\mu(x) = \varphi(x) \partial^\mu f(x)$$

(In writing this equation I have assumed that f is infinitesimal and neglected the terms $\sim f^2$). The situation meets the conditions of the Noether's theorem. It follows that every solution f of the KG equation generates a conserved current J_f^μ , such that

$$\partial_\mu J_f^\mu = 0 \quad \text{on shell.}$$

Explicitly

$$J_f^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} f - \tilde{J}_f^\mu = \partial^\mu \varphi f - \varphi \partial^\mu f .$$

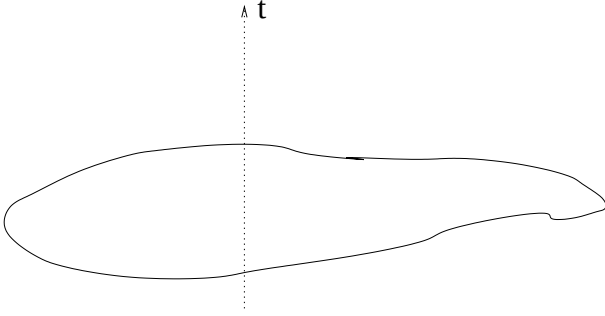
The corresponding Integrals of Motion are (at a time t)

$$A_f = \int d^3\mathbf{x} \left(\dot{\varphi}(\mathbf{x}) f(\mathbf{x}) - \varphi(\mathbf{x}) \dot{f}(\mathbf{x}) \right) = \int d^3\mathbf{x} \left(\pi(\mathbf{x}) f(\mathbf{x}) - \varphi(\mathbf{x}) \dot{f}(\mathbf{x}) \right) , \quad (2.1)$$

where $\pi(\mathbf{x}) = \dot{\varphi}(\mathbf{x})$ is the canonical momentum conjugate to $\varphi(\mathbf{x})$. The integrals A_f are Lorentz scalars. This becomes explicit if one rewrites their definition in covariant form

$$A_f = \int_\Sigma d\Sigma_\mu \left(\partial^\mu \varphi f - \varphi \partial^\mu f \right) ,$$

where Σ is any space-like 3-surface; because of the continuity equation the currents J_f^μ obey, the above integrals do not depend on specific choice of Σ .



In classical theory one straightforwardly evaluates the Poisson brackets

$$\{A_f, A_g\} = \int d^3\mathbf{x} \left(\dot{f}(\mathbf{x}) g(\mathbf{x}) - f(\mathbf{x}) \dot{g}(\mathbf{x}) \right)$$

(This expression also can be written in the covariant form, in terms of the integral over arbitrary Σ).

In quantum theory we should replace, as usual,

$$\{A, B\} \rightarrow i [\hat{A}, \hat{B}] ,$$

so that

$$[\hat{A}_f, \hat{A}_g] = -i \int d^3\mathbf{x} \left(\dot{f}(\mathbf{x}) g(\mathbf{x}) - f(\mathbf{x}) \dot{g}(\mathbf{x}) \right) .$$

It is convenient to use standard plane-wave solutions of the Klein-Gordon equation

$$f_{\mathbf{p}} = e^{i\omega_{\mathbf{p}} t - i\mathbf{p}\mathbf{x}} , \quad f_{\mathbf{p}}^* = e^{-i\omega_{\mathbf{p}} t + i\mathbf{p}\mathbf{x}} ,$$

where

$$\omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} .$$

Denoting

$$\hat{A}_{\mathbf{p}} = \hat{A}_{f_{\mathbf{p}}} , \quad \hat{A}_{\mathbf{p}}^\dagger = \hat{A}_{f_{\mathbf{p}}^*} ,$$

we find the commutation relations

$$[\hat{A}_{\mathbf{p}}, \hat{A}_{\mathbf{p}'}^\dagger] = (2\pi)^3 2\omega_{\mathbf{p}} \delta^{(3)}(\mathbf{p} - \mathbf{p}') ;$$

$$[\hat{A}_{\mathbf{p}}, \hat{A}_{\mathbf{p}'}] = 0 ; \quad [\hat{A}_{\mathbf{p}}^\dagger, \hat{A}_{\mathbf{p}'}^\dagger] = 0 .$$

These are the *creation and annihilation operators*. The above operators $\hat{A}_{\mathbf{p}}$ and $\hat{A}_{\mathbf{p}}^\dagger$ differ in normalization from $a_{\mathbf{p}}$ and $a_{\mathbf{p}}^\dagger$ used in PS (see **Sect.2.3**)

$$\hat{A}_{\mathbf{p}} = \sqrt{2\omega_{\mathbf{p}}} a_{\mathbf{p}} ;$$

$$\hat{A}_{\mathbf{p}}^\dagger = \sqrt{2\omega_{\mathbf{p}}} a_{\mathbf{p}}^\dagger .$$

Advantage of $A, A^\dagger$ is their Lorentz invariance.

The Hamiltonian can be expressed through the creation and annihilation operators as

$$\hat{H} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{4} \left(\hat{A}_{\mathbf{p}}^\dagger \hat{A}_{\mathbf{p}} + \hat{A}_{\mathbf{p}} \hat{A}_{\mathbf{p}}^\dagger \right), \quad (2.2)$$

which indeed is the Hamiltonian of a collection of harmonic oscillators, one oscillator per each wave-vector $\mathbf{p}$, the associated frequency being $\omega_{\mathbf{p}}$. It is straightforward to check that $\hat{A}_{\mathbf{p}}^\dagger$ rises (and $\hat{A}_{\mathbf{p}}$ lowers) the energy by the amount $\omega_{\mathbf{p}}$, namely

$$[\hat{H}, \hat{A}_{\mathbf{p}}^\dagger] = \omega_{\mathbf{p}} \hat{A}_{\mathbf{p}}^\dagger ,$$

$$[\hat{H}, \hat{A}_{\mathbf{p}}] = -\omega_{\mathbf{p}} \hat{A}_{\mathbf{p}} .$$

Similarly, for the momentum

$$\hat{\mathbf{P}} = - \int d^3\mathbf{x} \hat{\pi}(\mathbf{x}) \nabla \hat{\varphi}(\mathbf{x})$$

we have

$$\hat{\mathbf{P}} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\mathbf{p}}{4\omega_{\mathbf{p}}} \left(\hat{A}_{\mathbf{p}}^\dagger \hat{A}_{\mathbf{p}} + \hat{A}_{\mathbf{p}} \hat{A}_{\mathbf{p}}^\dagger \right) .$$

Hence

$$[\hat{\mathbf{P}}, \hat{A}_{\mathbf{p}}^\dagger] = \mathbf{p} \hat{A}_{\mathbf{p}}^\dagger ,$$

$$[\hat{\mathbf{P}}, \hat{A}_{\mathbf{p}}] = -\mathbf{p} \hat{A}_{\mathbf{p}} ,$$

i.e. $\hat{A}_{\mathbf{p}}^\dagger$ adds $\mathbf{p}$ to the total momentum.

To find a physically acceptable solution of quantum KG theory it remains to find appropriate representation of the above commutation relations in a Hilbert space $\mathcal{H}$. Classically, the Hamiltonian (2.1) is non-negative defined. Correspondingly,

under physically meaningful quantization the quantum Hamiltonian (2.2) must be bounded from below. Recall from the theory of harmonic oscillator that in order to satisfy this property one has to assume that $\mathcal{H}$ contains the ground state $|0\rangle$ such that

$$\hat{A}_{\mathbf{p}} |0\rangle = 0 \quad \text{for all } \mathbf{p}.$$

Just as in the harmonic oscillator, by applying the operators $\hat{A}_{\mathbf{p}}^\dagger$ to $|0\rangle$ one generates the complete space of normalizable states. In QFT this is referred as the *Fock space*,

$$\mathcal{H}_{\text{KG}} = \text{Fock space} = \text{Span} \left\{ \hat{A}_{\mathbf{p}_1}^\dagger \hat{A}_{\mathbf{p}_2}^\dagger \cdots \hat{A}_{\mathbf{p}_N}^\dagger |0\rangle \right\}.$$

The state $|0\rangle$ is called *vacuum* state, or *Fock vacuum*. Note that the above basis already solves the diagonalization problem for $\hat{H}$: these vectors are the eigenstates of both $\hat{H}$ and $\hat{\mathbf{P}}$, with the eigenvalues

$$E = \omega_{\mathbf{p}_1} + \omega_{\mathbf{p}_2} + \cdots + \omega_{\mathbf{p}_N} + E_0,$$

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2 + \cdots + \mathbf{p}_N + \mathbf{P}_0,$$

where E_0 and $\mathbf{P}_0$ are the eigenvalues associated with the vacuum state $|0\rangle$,

$$\hat{H} |0\rangle = E_0 |0\rangle, \quad \hat{\mathbf{P}} |0\rangle = \mathbf{P}_0 |0\rangle.$$

The basic vectors $\hat{A}_{\mathbf{p}_1}^\dagger \hat{A}_{\mathbf{p}_2}^\dagger \cdots \hat{A}_{\mathbf{p}_N}^\dagger |0\rangle$ are interpreted as the states of N identical (Bose) free particles, with the momenta $\mathbf{p}_1, \dots, \mathbf{p}_N$.

The above expressions for the Hamiltonian can be rewritten as

$$\hat{H} = E_0 + \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2} \hat{A}_{\mathbf{p}}^\dagger \hat{A}_{\mathbf{p}},$$

with

$$E_0 = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{4} [\hat{A}_{\mathbf{p}}, \hat{A}_{\mathbf{p}}^\dagger],$$

which of course is the sum of the zero-point energies of all the constituent oscillators.

There are two evident problems with the last expression:

(i) It is not zero, and moreover it contains seemingly meaningless factor $\delta^{(3)}(\mathbf{0})$ (recall that $[\hat{A}_{\mathbf{p}}, \hat{A}_{\mathbf{p}'}^\dagger] = 2\omega_{\mathbf{p}} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}')$).

(ii) The remaining integral over $\mathbf{p}$ diverges at large $\mathbf{p}$.

Note that these problem don't seem to be as bad for $\mathbf{P}_0$, where we have

$$\mathbf{P}_0 = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\mathbf{p}}{4\omega_{\mathbf{p}}} [\hat{A}_{\mathbf{p}}, \hat{A}_{\mathbf{p}}^\dagger] = 0,$$

since the contributions of $\mathbf{p}$ and $-\mathbf{p}$ cancel pairwise. This is good, as we do not want the vacuum to have any net momentum. But this seems to create another problem, if one thinks that the energy and momentum should be the components of 4-vector $(E_0, \mathbf{P}_0)$. If $\mathbf{P}_0 = 0$ and $E_0 \neq 0$, it seems the Lorentz invariance is broken.

To sort this part of the problem out, let us recall that we are dealing with a system in which the degrees of freedom $\varphi(\mathbf{x})$ are attached to all points of the space. For such system one expects that its vacuum energy E_0 , whatever it is, must be proportional to the volume of the space,

$$E_0 = \varepsilon V^{(3)},$$

where ε is the vacuum energy density. This answers the question how to interpret the $\delta^{(3)}(\mathbf{0})$ factor: Recall that

$$(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') = \int d^3\mathbf{x} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}},$$

and so the above nonsense factor in fact is

$$(2\pi)^3 \delta^{(3)}(\mathbf{0}) = \int d^3\mathbf{x} \rightarrow V^{(3)}.$$

This also solves the problem of Lorentz invariance. The energy density ε transforms as 00 component of 4-tensor (the energy-momentum tensor),

$$\varepsilon = \langle 0 | T^{00} | 0 \rangle.$$

The Lorentz-invariance of the vacuum state $|0\rangle$ only requires that

$$\langle 0 | T^{\mu\nu} | 0 \rangle = \varepsilon_0 \eta^{\mu\nu}, \quad \eta^{\mu\nu} = \text{diag}(+1, -1, -1, -1),$$

and we soon will have techniques to check that this equation indeed holds in the KG theory. It is OK for a Lorentz-invariant state to have nonzero energy density.

Still, something has to be done about the badly divergent integral for this energy density,

$$\varepsilon_0 = \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \omega_{\mathbf{p}}.$$

The integral diverges at large momenta, i.e. the divergent contribution comes from small length scales. This is not too surprising, since our idea of the field theory was to associate a degree of freedom with every point of the space, and there are continuously many points in each finite part of it. One could assume that perhaps some microscopic physics takes care of the problem, replacing the above mathematically meaningless expression by the “cutoff” integral

$$\varepsilon_0 = \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \omega_{\mathbf{p}} \Phi\left(\frac{\mathbf{p}^2}{\Lambda^2}\right).$$

where the “cutoff” factor $\Phi(\xi)$ is close to 1 when $\xi \ll 1$ and decays fast at large ξ , and Λ represents the energy scale beyond which the unknown microscopic physics becomes essential.

Whereas introducing the cutoff rids us of mathematically meaningless expression, needless to say it does not solve the problem of vacuum energy density, which now is finite but completely undetermined. In many practical cases one can simply ignore the problem. Large class of physical quantities are only sensitive to differences of energies, and in such problems the vacuum energy is invisible. But in fact the vacuum energy is measurable, in principle. Generally, if we want to be able to apply QFT to physics, we need to know what happens when we couple it to gravity, and gravity is very sensitive to the energy density. Even without gravity, suppose we can confine our system to finite part of the space, say inside a box. Then we want to know what forces it exerts upon the wall of that box. The last situation can actually be realized in laboratory, and it leads to a measurable effect - the so called Casimir effect. We are going to discuss it next, but let me make one more remark, in order to clarify the nature of the problem.

When a divergent integral appears in otherwise sensible calculations, it usually signals some ambiguity in the theory. Indeed, there is an ambiguity in the definition of the KG theory. One can add a constant term to the Lagrangian density,

$$\mathcal{L} \rightarrow \mathcal{L} + \text{Const}.$$

This extra term is fully consistent with both relativistic symmetry and locality. Of course this modification is completely irrelevant in the classical theory, since it does not affect the equations of motion. But the Lagrangian density will pass this term to the energy-momentum tensor, and finally to the Hamiltonian,

$$\hat{H} \rightarrow \hat{H} - \text{Const} \int d^3\mathbf{x}.$$

One can now “absorb” the vacuum energy divergence into this constant. We write

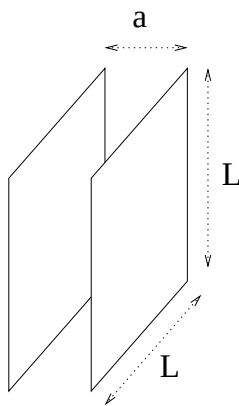
$$\text{Const} = \varepsilon_0 - \varepsilon ,$$

where ε_0 is the above ugly integral, and ε is a finite constant (of the dimension $[\text{Mass}]^4$). The result is that the divergent integral disappears, but at the same time we learn that the quantum KG theory in fact has not one but two parameters, m^2 and ε . This extra term we added to the action is the simplest example of what we will call the *counterterms*.

3 Lecture 3

Casimir Effect

To understand possible manifestations of the vacuum energy, let us now briefly discuss the *Casimir Effect*. Originally, the Casimir effect refers to the situation where we have two parallel conducting plates, of large size $L \times L$, separated by a distance a .



Due to the quantum nature of the electromagnetic field in between the plates, more precisely, due to the effect of the plates on the zero-point energy of quantum electromagnetic field, the plates are attracted with the force

$$\frac{\text{Force}}{L^2} = -\hbar c \frac{\pi^2}{240 a^4},$$

where it is assumed that

$$L \gg a \gg \text{atomic distances.}$$

Casimir effect in KG theory

Let us study the nature of this effect using our scalar field theory, the KG theory, instead of the electrodynamics; the result is not going to be very different. Since the real photons are massless (and to make calculations simpler), we will assume that $m^2 = 0$. In macroscopic electromagnetic theory (valid at the scales $\gg$ atomic distances) the presence of the conducting plates is taken into account by imposing the boundary conditions $\mathbf{E}_{\parallel} = 0$, $\mathbf{B} = 0$ at the surface of the conductor.

To mimic this situation we consider instead the KG field in the presence of two plates with the boundary condition

$$\varphi(\mathbf{x}) = 0 \quad \text{at the surface.}$$

Let us concentrate attention on the field between the plates. In order to take into account the boundary conditions, in this case we must use, instead of the functions $f_{\mathbf{p}}(x)$ and $f_{\mathbf{p}}^*(x)$ above, the following complete set of solutions

$$f_{\mathbf{p}_{\parallel},n}(\mathbf{x}, \mathbf{t}) = \sqrt{\frac{2}{a}} e^{i\omega_{\mathbf{p},n} \mathbf{t} - i\mathbf{p}_{\parallel} \cdot \mathbf{x}_{\parallel}} \sin\left(\frac{\pi \mathbf{n}}{a} \mathbf{x}_{\perp}\right),$$

and the corresponding complex-conjugated functions $f_{\mathbf{p}_{\parallel},n}^*$. Here $\mathbf{x}_{\parallel}$ and $x_{\perp}$ are the components of $\mathbf{x}$ parallel and perpendicular to the plates, $\mathbf{x} = (\mathbf{x}_{\parallel}, x_{\perp})$, and

$$\omega_{\mathbf{p}_{\parallel},n} = \sqrt{\mathbf{p}_{\parallel}^2 + \frac{\pi^2 n^2}{a^2}}$$

with positive integer and n ,

$$n = 1, 2, 3, \dots$$

These functions solve the KG equation with $m^2 = 0$ as well as the boundary conditions

$$f_{\mathbf{p}_{\parallel},n}|_{x_{\perp}=0} = 0, \quad \text{and} \quad f_{\mathbf{p}_{\parallel},n}|_{x_{\perp}=a} = 0.$$

Exactly as we did in the infinite-volume case, we now introduce the operators

$$\hat{A}_{\mathbf{p}_{\parallel},n} = \int_0^a dx_{\perp} \int d^2\mathbf{x}_{\parallel} [\hat{\pi} f_{\mathbf{p}_{\parallel},n} - \hat{\varphi} \dot{f}_{\mathbf{p}_{\parallel},n}],$$

and similarly for $\hat{A}_{\mathbf{p}_{\parallel},n}^{\dagger}$, with $f_{\mathbf{p}_{\parallel},n}^*$ replacing $f_{\mathbf{p}_{\parallel},n}$. It is not difficult then to show that

$$[\hat{A}_{\mathbf{p}_{\parallel},n}, \hat{A}_{\mathbf{p}'_{\parallel},n'}^{\dagger}] = 2\omega_{\mathbf{p}_{\parallel},n} \delta_{nn'} (2\pi)^2 \delta(\mathbf{p}_{\parallel} - \mathbf{p}'_{\parallel}),$$

and to compute the Hamiltonian of the part of the system residing between the plates,

$$\hat{H} = \sum_{n=1}^{\infty} \int \frac{d^2\mathbf{p}_{\parallel}}{(2\pi)^2} \frac{1}{4} \left(\hat{A}_{\mathbf{p}_{\parallel},n}^{\dagger} \hat{A}_{\mathbf{p}_{\parallel},n} + \hat{A}_{\mathbf{p}_{\parallel},n} \hat{A}_{\mathbf{p}_{\parallel},n}^{\dagger} \right) + a L^2 (\varepsilon - \varepsilon_0),$$

where the last term comes from vacuum energy counterterm

$$(\varepsilon - \varepsilon_0) \int_0^a dx_\perp \int d^2 \mathbf{x}_\parallel = a L^2 (\varepsilon - \varepsilon_0).$$

We still have to demand that all the operators $\hat{A}_{\mathbf{p}_\parallel, n}$ nullify the ground state, since otherwise $\hat{H}$ is unbounded from below. Therefore the vacuum eigenvalue of this Hamiltonian is

$$E_{\text{vac}} = \frac{L^2}{2} \sum_{n=1}^{\infty} \int \frac{d^2 \mathbf{p}_\parallel}{(2\pi)^2} \sqrt{\mathbf{p}_\parallel^2 + \frac{\pi^2 n^2}{a^2}} \Phi\left(\frac{\mathbf{p}_\parallel^2 + \pi^2 n^2/a^2}{\Lambda^2}\right) - a L^2 \varepsilon_0 + a L^2 \varepsilon,$$

where I have introduced again the cutoff factor Φ - without it this expression is as divergent as it was in the absence of the boundaries.

Now, the term $-a L^2 \varepsilon_0$ is the divergent part of the vacuum energy counterterm. It is

$$-a L^2 \varepsilon_0 = -\frac{L^2}{2} a \int \frac{dp_\perp}{(2\pi)} \int \frac{d\mathbf{p}_\parallel}{(2\pi)^2} \sqrt{\mathbf{p}_\parallel^2 + p_\perp^2} \Phi\left(\frac{\mathbf{p}_\parallel^2 + p_\perp^2}{\Lambda^2}\right),$$

where I have separated the $dp_\perp$ and $d^2 \mathbf{p}_\parallel$ parts of the integral over $d^3 \mathbf{p}$. After the change of variables

$$p_\perp = \pi \tau / a$$

it takes the form

$$-\frac{L^2}{2} \int_0^\infty d\tau \int \frac{d^2 \mathbf{p}_\parallel}{(2\pi)^2} \sqrt{\mathbf{p}_\parallel^2 + \frac{\pi^2 \tau^2}{a^2}} \Phi\left(\frac{\mathbf{p}_\parallel^2 + \pi^2 \tau^2/a^2}{\Lambda^2}\right).$$

We see that the interesting part of the ground-state energy (rather, of the energy per unit area of the plates) can be written as

$$\frac{E}{L^2} - a \varepsilon = \int \frac{d^2 \mathbf{p}_\parallel}{(2\pi)^2} \left[\sum_{n=1}^{\infty} F_{\mathbf{p}_\parallel}(n) - \int_0^\infty F_{\mathbf{p}_\parallel}(\tau) d\tau \right],$$

where

$$F_{\mathbf{p}_\parallel}(\tau) = \frac{1}{2} \sqrt{\mathbf{p}_\parallel^2 + \pi^2 \tau^2/a^2} \Phi\left(\frac{\mathbf{p}_\parallel^2 + \pi^2 \tau^2/a^2}{\Lambda^2}\right).$$

The $\sum F(n) - \int F(\tau)$ expression above can be handled with the help of the Euler-McLaurin summation formula (see Appendix)

$$\sum_{n=1}^{\infty} F(n) - \int_0^\infty F(\tau) d\tau = -\frac{1}{2} F(0) + i \int_0^\infty \frac{F(it+0) - F(-it+0)}{e^{2\pi t} - 1} dt. \quad (3.1)$$

which is valid provides $F(\tau)$ is analytic in the right half-plane of complex τ . The first term

$$-\frac{1}{2} F_{\mathbf{p}_{\parallel}}(0) = -\frac{1}{4} \int \frac{d^2 \mathbf{p}_{\parallel}}{(2\pi)^2} \sqrt{\mathbf{p}_{\parallel}^2} \Phi(\mathbf{p}_{\parallel}^2/\Lambda^2).$$

does not depend on the separation a ; it is interpreted as the surface energy. It diverges, again signaling ambiguity. When there is a surface, a constant surface term can be added to the action. We ignore this term since, being independent of a it does not contribute to the force.

The interesting contribution is extracted from the second term, which involves

$$i \int_0^\infty \frac{F(it+0) - F(-it+0)}{e^{2\pi t} - 1} dt$$

The the shift ”+0” in the arguments in (EM) is important since the function $F_{\mathbf{p}_{\parallel}}(\tau)$ has square-root brunching points at the imaginary axis. Note that if it was regular, the integral would vanish in view of the obvious $\tau \rightarrow -\tau$ symmetry of $F_{\mathbf{p}_{\parallel}}(\tau)$. In fact, one can take advantage of this symmetry to write the integrand as

$$i \int_0^\infty \frac{F(it+0) - F(it-0)}{e^{2\pi t} - 1} dt = -2 \int_{|\mathbf{p}_{\parallel}|a/\pi}^\infty \frac{\Im F(it+0)}{e^{2\pi t} - 1} dt,$$

where the last form reflects the fact that the function $F_{\mathbf{p}_{\parallel}}(\tau)$ has the branch cut from $i|\mathbf{p}|a/\pi$ to $i\infty$ (and from $-i|\mathbf{p}|a/\pi$ to $-i\infty$). We have, at $|t| > a|p|/\pi$

$$\Im F_{\mathbf{p}_{\parallel}}(it+0) = \frac{1}{2} \sqrt{\frac{\pi^2 t^2}{a^2} - \mathbf{p}_{\parallel}^2} \Phi\left(\frac{\mathbf{p}_{\parallel}^2 - \pi^2 t^2/a^2}{\Lambda^2}\right) := \frac{1}{2} G_{\mathbf{p}_{\parallel}}(t).$$

Note that the factor $(e^{2\pi t} - 1)^{-1}$ makes this integral fast convergent at large t , even without the cutoff factor Φ . At large $\Lambda \gg a^{-1}$ this factor has no effect, and can be dropped,

$$G_{\mathbf{p}_{\parallel}}(t) \rightarrow \sqrt{\frac{\pi^2 t^2}{a^2} - \mathbf{p}_{\parallel}^2}.$$

The interesting contribution to the vacuum energy then is

$$- \int \frac{dt}{e^{2\pi t} - 1} \int_{|\mathbf{p}_{\parallel}| \leq \pi t/a} \frac{d^2 \mathbf{p}_{\parallel}}{(2\pi)^2} \sqrt{\frac{\pi^2 t^2}{a^2} - \mathbf{p}_{\parallel}^2}.$$

where I have interchanged the order in which the integrals are taken. The integral over $\mathbf{p}_{\parallel}$ is evaluated in closed form

$$\int_0^{\frac{\pi^2 t^2}{a^2}} \frac{dp^2}{4\pi} \sqrt{\frac{\pi^2 t^2}{a^2} - p^2} = \frac{1}{6\pi} \left(\frac{\pi t}{a}\right)^3,$$

The contribution to the energy is

$$-\frac{\pi^2}{6a^3} \int_0^\infty \frac{t^3 dt}{e^{2\pi t} - 1} = -\frac{\pi^2}{6a^3} \frac{\Gamma(4) \zeta(4)}{(2\pi)^4}$$

where I have used

$$\int_0^\infty \frac{x^{s-1} dx}{e^x - 1} = \Gamma(s) \zeta(s).$$

($\zeta(4) = \pi^4/90$). Thus we find

$$\frac{E}{L^2} = a\varepsilon - \frac{1}{2} F(0) - \frac{\pi^2}{1440a^3}$$

The contribution $a\varepsilon$ is not too interesting - there is the same term in the vacuum energy density outside the space between the plates, and hence this term does not lead to any force. The interesting term $-\pi^2/1440a^3$ shows that the plates attract; the force (per unit plate area) is

$$\frac{\text{Force}}{L^2} = \frac{\pi^2 \hbar c}{480a^4}.$$

Let me stress that this result is completely independent of the form of auxiliary cutoff factor Φ - this is exactly what we want from quantum field theory. Another useful observation is that in order to obtain this result we don't really need to send Λ to Planck's energies or something, it suffices to have $a^2 \gg 1/\Lambda^2$.

In the real case of electromagnetic field the calculations can be done in very similar way. The resulting force is twice as large, due to the fact that photon field has two degrees of freedom for each $\mathbf{p}$, as the photon has two polarization states.

Few words about the a -independent term

$$-\frac{1}{2} F(0) = -\frac{1}{16\pi} \int_0^\infty d\xi \sqrt{\xi} \Phi(\xi/\Lambda^2).$$

It does not contribute to the force, but still needs some interpretation. It is divergent, and it is easy to design special counter-term to absorb this divergence. But it is instructive to think about real case of electromagnetic field and real conducting plates made of atoms, conducting electrons, and all that. In this case the cutoff represents the limit of validity of the macroscopic electrodynamics, and the cutoff energy Λ in this expression should be taken around atomic energy scale. Having sufficient knowledge about the microscopic structure of the conductor we could in

principle calculate the cutoff factor and hence this contribution in terms of the atomic parameters. The nice feature of the above result is in that it predicts some attractive force between the plates which is completely independent of all these microscopic details. This points to another area where quantum field theory applies, when it is not that we can't know the short-distance physics, rather we don't want to know. In this area the quantum field theory describes universal, i.e. not too sensitive to microscopic details, properties of matter.

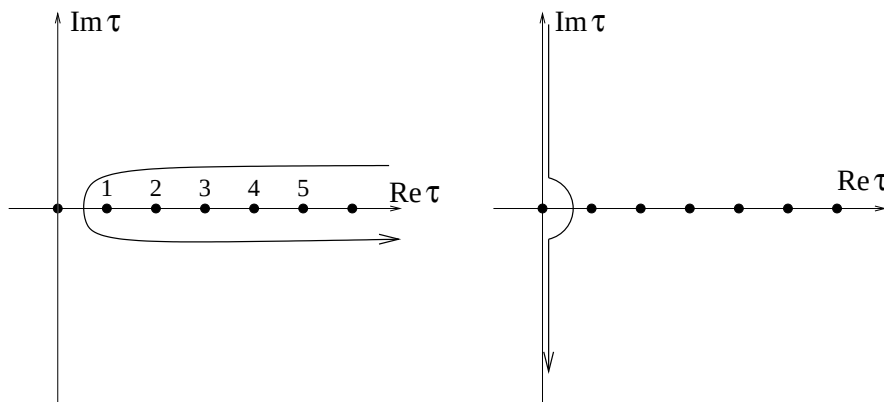
Appendix: Euler-Maclaurin formula

Let $F(\tau)$ be a function analytic in the right half-plane, and decaying at $\Re \tau \rightarrow +\infty$ sufficiently fast, so that the integral $\int_0^\infty f(\tau) d\tau$ converges.

Start with the integral

$$I = \int_{C_0} \frac{F(\tau) d\tau}{e^{2\pi i \tau} - 1},$$

where the integration is over the contour C which encloses all positive integer points $\tau = 1, 2, 3, \dots$, as shown in the left figure.



The integrand has poles at the integer points $\tau = n$, with the residues $F(n)/2\pi i$. Therefore

$$I = \sum_{n=1}^{\infty} F(n).$$

On the other hand, the integral can be split into two parts

$$I = \int_{C_0} = \int_{C_+} + \int_{C_-} = I_+ + I_- ,$$

where C_+ is the part of the contour C_0 that lays above the real axis, and C_- is the part below the real axis. Note that the factor

$$\frac{1}{e^{2\pi i\tau} - 1}$$

decays fast into the lower half-plane, hence the contour C_- can be rotated into the lower half-plane to go along the contour $\tilde{C}_-$,

$$I_- = \int_{\tilde{C}_-} \frac{F(\tau) d\tau}{e^{2\pi i\tau} - 1} = -i \int_{\epsilon}^{\infty} \frac{F(-it + 0) dt}{e^{2\pi t} - 1} + \int_{C'_-} \frac{F(\tau) d\tau}{e^{2\pi i\tau} - 1},$$

where for the straight part of C_- I have changed

$$\tau = -it,$$

and C'_- is the small quarter-circle below the real axis, seen in the right part of the figure. It is easy to check that in the limit $\epsilon \rightarrow 0$ this part of the integral yields $-F(0)/4$.

On the other hand the factor $(e^{2\pi i\tau} - 1)^{-1}$ does not decay into the upper half plane, but one can write

$$\frac{1}{e^{2\pi i\tau} - 1} = -1 - \frac{1}{e^{-2\pi i\tau} - 1},$$

where the second term do decay into the upper half-plane, so that

$$I_+ = - \int_{C_+} F(\tau) d\tau + \int_{\tilde{C}_+} \frac{F(\tau) d\tau}{e^{-2\pi i\tau} - 1} = \int_{\epsilon}^{\infty} F(\tau) d\tau + i \int_{\epsilon}^{\infty} \frac{F(it + 0) dt}{e^{2\pi t} - 1} + \int_{C'_+},$$

where the second term in the last form represents the contribution of the straight part of C_+ (with $\tau = it + 0$), and the last term is the contribution of the small quarter-circle above the real axis; again, it is easy to check that this last contribution is $-F(0)/4$ in the limit $\epsilon \rightarrow 0$. Adding I_+ to I_- , and taking the limit $\epsilon = 0$, yields (3.1).

4 Lecture 4

Quantum Klein-Gordon theory (Continued)

The space of states $\mathcal{H}$ of the quantum KG theory is the Fock space, defined as the linear envelope of the basic states

$$|\mathbf{p}_1, \dots, \mathbf{p}_N\rangle = \hat{A}_{\mathbf{p}_1}^\dagger \dots \hat{A}_{\mathbf{p}_N}^\dagger |0\rangle,$$

where the vacuum $|0\rangle$ satisfies

$$\hat{A}_{\mathbf{p}} |0\rangle = 0 \quad \text{for all } \mathbf{p}.$$

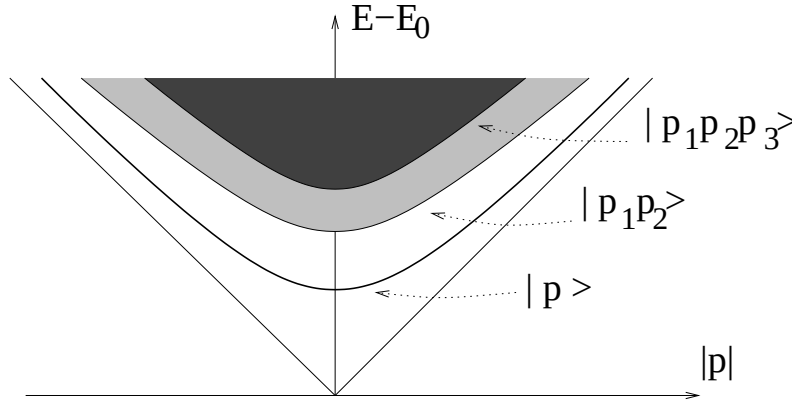
Here I have returned to the default case of the infinite Minkowski space-time, so that the 3-momenta $\mathbf{p}_i$ may take continuous unrestricted values.

The states $|\mathbf{p}_1, \dots, \mathbf{p}_N\rangle$ are interpreted as N -particle states. Their scalar products follow from the commutation relations $[\hat{A}_{\mathbf{p}}, \hat{A}_{\mathbf{p}'}^\dagger] = 2\omega_{\mathbf{p}} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}')$. In particular, for the one-particle states

$$\langle \mathbf{p} | \mathbf{p}' \rangle = 2\omega_{\mathbf{p}} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}').$$

Since the operators $\hat{A}_{\mathbf{p}}^\dagger$ are Lorentz invariant (only the momentum $\mathbf{p}$ transforms under the Lorentz transformations), the particles have zero spin. Since the creation operators $\hat{A}_{\mathbf{p}}^\dagger$ commute, the N -particle states are symmetric with respect to permutations of the momenta $\mathbf{p}_i$ - the particles are bosons.

The spectrum of the energy-momentum operator $(\hat{H} - E_0, \hat{\mathbf{P}})$ is the composition of spectra of the N -particle states. In the space of vectors $(E - E_0, \mathbf{P})$ the vacuum resides at the origin of coordinates, the one particle states form the mass-shell hyperboloid $(E - E_0)^2 - \mathbf{P}^2 = m^2$, the two-particle states fill the inside of the hyperboloid $(E - E_0)^2 - \mathbf{P}^2 = 4m^2$, etc.



Correspondingly, the spectrum of the operator $\hat{M}^2 = (\hat{H} - E_0)^2 - \hat{\mathbf{P}}^2$ has two discrete components, 0 and m^2 , and continuous part from $4m^2$ to infinity.

Klein-Gordon propagator

By their definition, the creation and annihilation operators $\hat{A}_{\mathbf{p}}^\dagger, \hat{A}_{\mathbf{p}}$ are the Fourier components of the local field operators $\hat{\varphi}(\mathbf{x})$ and $\hat{\pi}(\mathbf{x})$,

$$\hat{\varphi}(\mathbf{x}) = \int d\mu(\mathbf{p}) [\hat{A}_{\mathbf{p}} + \hat{A}_{-\mathbf{p}}^\dagger] e^{i\mathbf{p}\mathbf{x}},$$

$$\hat{\pi}(\mathbf{x}) = \int d\mu(\mathbf{p}) i\omega_{\mathbf{p}} [-\hat{A}_{\mathbf{p}} + \hat{A}_{-\mathbf{p}}^\dagger] e^{i\mathbf{p}\mathbf{x}},$$

where I have introduced special notation for the Lorentz-invariant measure on the mass shell,

$$d\mu(\mathbf{p}) = \frac{1}{2\omega_{\mathbf{p}}} \frac{d^3\mathbf{p}}{(2\pi)^3}.$$

Exercise: Check that the measure $d\mu(\mathbf{p})$ is Lorentz-invariant. An instructive way to do that is to check the identity

$$d\mu(\mathbf{p}) = \int_{p^0 > 0} dp^0 \delta(p^\mu p_\mu - m^2),$$

i.e. we have

$$\int d\mu(\mathbf{p}) F(\mathbf{p}, \omega_{\mathbf{p}}) = \int_{p^0 > 0} \frac{d^4 p}{(2\pi)^4} 2\pi \delta(p^\mu p_\mu - m^2) F(\mathbf{p}, p^0).$$

The operators $\varphi(\mathbf{x})$ and $\pi(\mathbf{x})$ are the "Schroedinger", or the "equal time" field operators. As defined, they obey the canonical commutation relations

$$[\hat{\pi}(\mathbf{x}), \hat{\varphi}(\mathbf{x}')] = -i \delta^{(3)}(\mathbf{x} - \mathbf{x}').$$

As is common in quantum mechanics, it is often useful to trade the time evolution of the states for the time evolution of the operators. This is known as the Heisenberg picture. In our case, introduce the Heisenberg field operator

$$\hat{\varphi}(x) = \hat{\varphi}(\mathbf{x}, t) = e^{i\hat{H}t} \hat{\varphi}(\mathbf{x}) e^{-i\hat{H}t}, \quad x = (\mathbf{x}, t).$$

In terms of the creation and annihilation operators it is

$$\hat{\varphi}(x) = \int d\mu(\mathbf{p}) [\hat{A}_{\mathbf{p}} f_{\mathbf{p}}^*(x) + \hat{A}_{\mathbf{p}}^\dagger f_{\mathbf{p}}(x)] ,$$

where as before $f_{\mathbf{p}}(x) = e^{i\omega_{\mathbf{p}}t - i\mathbf{p}\mathbf{x}}$. Let us compute the commutator $[\hat{\varphi}(x), \hat{\varphi}(x')]$. Then

$$\hat{\varphi}(\mathbf{x}) = \hat{\varphi}(\mathbf{x}, \mathbf{t} = \mathbf{0}) , \quad \hat{\pi}(\mathbf{x}) = \partial_{\mathbf{t}} \hat{\varphi}(\mathbf{x}, \mathbf{t}) |_{\mathbf{t}=\mathbf{0}} .$$

Commutators

It is interesting and instructive to see how the canonical commutators of the "Schroedinger" operators extend to the Heisenberg field operators $\hat{\varphi}(x)$. For this calculation, it is useful to write

$$\hat{\varphi}(x) = \hat{\varphi}_-(x) + \hat{\varphi}_+(x) ,$$

where

$$\hat{\varphi}_-(x) = \int d\mu(\mathbf{p}) \hat{A}_{\mathbf{p}} f_{\mathbf{p}}^*(x) , \quad \hat{\varphi}_+(x) = \int d\mu(\mathbf{p}) \hat{A}_{\mathbf{p}}^\dagger f_{\mathbf{p}}(x) ,$$

The operator $\hat{\varphi}_+(x)$ contains only the creation operators $\hat{A}_{\mathbf{p}}^\dagger$; it can be interpreted as an operator creating particle at the space-time point x ; likewise, $\hat{\varphi}_-(x)$ absorbs the particle at the space-time point x . Since $\hat{\varphi}_-(x)$ obviously commutes with $\hat{\varphi}_-(x')$, and the same is true for the $\hat{\varphi}_+$ component, we have

$$[\hat{\varphi}(x), \hat{\varphi}(x')] = D_-(x - x') - D_+(x - x') ,$$

where

$$\begin{aligned} D_-(x - x') &= [\hat{\varphi}_-(x), \hat{\varphi}_+(x')] , \\ D_+(x - x') &= [\hat{\varphi}_-(x'), \hat{\varphi}_+(x)] \equiv D_-(x' - x) . \end{aligned}$$

Explicitly

$$\begin{aligned} D_-(\mathbf{x}, t) &= \int d\mu(\mathbf{p}) e^{-i\omega_{\mathbf{p}}t + i\mathbf{p}\mathbf{x}} , \\ D_+(\mathbf{x}, t) &= \int d\mu(\mathbf{p}) e^{i\omega_{\mathbf{p}}t - i\mathbf{p}\mathbf{x}} , \end{aligned}$$

Some properties of these integrals are worth mentioning.

i) The integrals for $D_-(x)$ and $D_+(x)$ are Lorentz-invariant, i.e. if two 4-vectors x and x' are related by a Lorentz transformation, $x' = \Lambda x$, then $D(x) = D(x')$. Indeed, the integrals can be written in explicitly covariant form, say

$$D_-(x) = \int_{p^0 > 0} \frac{d^4 p}{(2\pi)^4} 2\pi \delta(p^\mu p_\mu - m^2) e^{-i p_\mu x^\mu} .$$

Then the change of the integration variables $p = \Lambda^{-1}p'$ verifies that $D_-(x) = D_-(\Lambda x)$. Similar representation exists for $D_+(x)$.

ii) The integrals are not absolutely convergent. It is usually convenient to compute such integrals as some limiting values of absolutely convergent integrals. In our case it can be done by considering the integrals at complex values of the time variable t . For instance, the above integral defining $D_-(\mathbf{x}, t)$ converges absolutely if t has **negative imaginary part** (remember that $\omega_{\mathbf{p}}$ is positive), therefore this integral defines the function $D_-(\mathbf{x}, t)$ of complex variable t which is analytic in the lower half-plane of the complex t -plane. Then, for real t we need to take the limit $\lim_{\epsilon \rightarrow -0} D_-(\mathbf{x}, t + i\epsilon)$ (this is usually written as $D_-(\mathbf{x}, t - i0)$). Likewise, the integral for D_+ defines the function $D_+(\mathbf{x}, t)$ of complex t , analytic in the upper half-plane, and for real t we take $D_+(\mathbf{x}, t + i0)$.

Let us consider the case of real t such that the 4-vector $x^\mu = (t, \mathbf{x})$ is space-like,

$$x^\mu x_\mu = t^2 - \mathbf{x}^2 < 0$$

(in this consideration we assume that $\mathbf{x}$ is real). In the complex t -plane we are looking at the segment

$$-|\mathbf{x}| < t < |\mathbf{x}|$$

of the real axis. It is easy to show that at this segment the we have

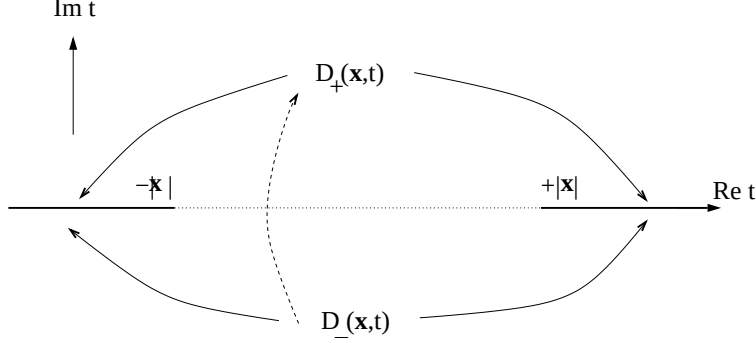
$$D_+(\mathbf{x}, t) = D_-(\mathbf{x}, t) \quad \text{for} \quad -|\mathbf{x}| < t < |\mathbf{x}|$$

It suffices to verify this identity at $t = 0$, since one can always transform to the frame where $t = 0$, as long as x^μ is space-like. For $t = 0$ we have

$$D_-(\mathbf{x}, 0) = \lim_{\epsilon \rightarrow +0} \int e^{-\omega_{\mathbf{p}} \epsilon + i\mathbf{p} \cdot \mathbf{x}} d\mu(\mathbf{p}) \quad \text{and} \quad D_+(\mathbf{x}, 0) = \lim_{\epsilon \rightarrow +0} \int e^{-\omega_{\mathbf{p}} \epsilon - i\mathbf{p} \cdot \mathbf{x}} d\mu(\mathbf{p}),$$

and the identity is established by the change of the integration variables $\mathbf{p} \rightarrow -\mathbf{p}$. Note that this argument also shows that $D_\pm(\mathbf{x}, t)$ return real values when t takes values within the above segment of the real axis.

It follows that the function $D_-(\mathbf{x}, t)$, which was originally defined (by out integral) in the lower half-plane of t , in fact can be analytically continued, through the above segment of the real axis, to the upper half-plane, and there it coincides with $D_+(\mathbf{x}, t)$ (defined, again, through its integral representation written above).



In other words, in fact there is a single function of complex t (I will denote it as $D(\mathbf{x}, t)$), analytic in both the upper and lower half-planes, including the segment $-|\mathbf{x}| < t < |\mathbf{x}|$ of the real axis, and $D_+(\mathbf{x}, t)$ ($D_-(\mathbf{x}, t)$) is given by its values in the upper (lower) half-plane. In particular, for real t

$$D_+(\mathbf{x}, t) = D(\mathbf{x}, t + i0),$$

$$D_-(\mathbf{x}, t) = D(\mathbf{x}, t - i0).$$

As we have seen, these two limits coincide when $-|\mathbf{x}| < t < |\mathbf{x}|$, but outside this segment they don't. That means the function $D(\mathbf{x}, t)$ has two branch cuts in the complex t -plane, from $-\infty$ to $-|\mathbf{x}|$, and from $|\mathbf{x}|$ to $+\infty$.

It is not difficult to identify the geometric meaning of the branching points $t = \pm|\mathbf{x}|$. They represent points on the future and past components of the light cone associated with the space-time point 0, that is the geometric place of all points which can be connected to 0 by light rays. At fixed $\mathbf{x}$ the real t line intersects the light cone at two points $\pm|\mathbf{x}|$. When t regarded as the complex variable, the intersection points become exactly the branching points in the complex t plane. At real t , when crossing the light cone one needs to choose the branch of the analytic function (similar to choosing a branch of $\sqrt{z}$ at negative z). The functions D_+ and D_- the two branches corresponding to two ways of going around the branching points in the complex t -plane.

Figure

Let us come back to the commutator $[\hat{\varphi}(x), \hat{\varphi}(0)] = D_-(x) - D_+(x)$. Here I have set $x' = 0$; in general case x has to be replaced by the difference $x - x'$. We see that the commutator is equal to the discontinuity

$$[\hat{\varphi}(\mathbf{x}, t), \hat{\varphi}(\mathbf{0}, 0)] = D(\mathbf{x}, t - i0) - D(\mathbf{x}, t + i0).$$

When the separation $x = (t, \mathbf{x})$ is space-like the discontinuity vanishes, and we conclude that

$$[\hat{\varphi}(x), \hat{\varphi}(x')] = 0 \quad \text{for all space-like separations} \quad x - x'.$$

This is important conclusion. Recall that in quantum mechanics commutator of two operators allows one to determine how, in a given state, the measurement of one observable can affects results of the measurements of another observable. Our calculation shows that the measurement at some space-time point x can not affect measurements at the point x' as long as the separation $x - x'$ is space-like. It expresses the **causality** of the quantum field theory. This is general requirement of quantum field theory called the **local commutativity**. It states that for any local fields $\mathcal{O}_1(x)$, $\mathcal{O}_2(x')$ we must have

$$[\hat{\mathcal{O}}_1(x), \hat{\mathcal{O}}_2(x')] = 0 \quad \text{for all space-like separations} \quad x - x'.$$

The discontinuities across the brunch cuts $|t| > |\mathbf{x}|$ do not vanish, so when the separation is time-like the commutator takes non-zero values. We see that the commutator has its support inside the light cone.

Consider now the expectation value $\langle 0 | \hat{\varphi}(x) \hat{\varphi}(x') | 0 \rangle$ of the product of two field operators. In the KG theory this expectation value can be expressed through the same analytic function $D(\mathbf{x}, t)$. Indeed, since $\hat{\varphi}_-(x) | 0 \rangle = 0$ and $\langle 0 | \hat{\varphi}_+(x) = 0$, we have

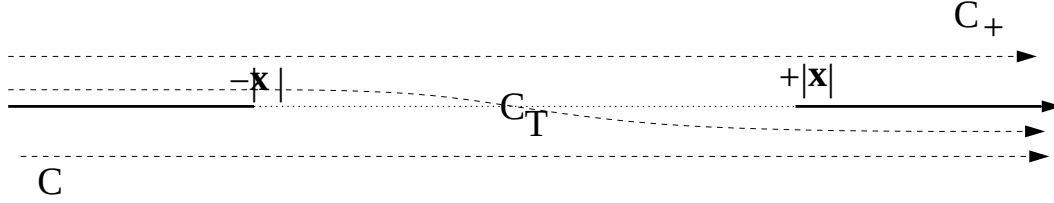
$$\langle 0 | \hat{\varphi}(x) \hat{\varphi}(x') | 0 \rangle = \langle 0 | \hat{\varphi}_-(x) \hat{\varphi}_+(x') | 0 \rangle = D_-(x - x') = D(\mathbf{x} - \mathbf{x}', t - t' - i0).$$

If we interchange the positions of the operators, by similar calculation

$$\langle 0 | \hat{\varphi}(x') \hat{\varphi}(x) | 0 \rangle = D_+(x - x') = D(\mathbf{x} - \mathbf{x}', t - t' + i0)$$

Thus, these expectation values are described by the function $D(\mathbf{x}, t)$ taken at the upper or lower edges of the light-cone branch cuts

$$\langle 0 | \hat{\varphi}(\mathbf{0}, 0) \hat{\varphi}(\mathbf{x}, t) | 0 \rangle$$



$$\langle 0 | \hat{\varphi}(\mathbf{x}, t) \hat{\varphi}(\mathbf{0}, 0) | 0 \rangle$$

Important role in our future studies will belong to the **time ordered** expectation values

$$D_F(x - x') = D_F(\mathbf{x} - \mathbf{x}', t - t') = \langle 0 | T(\hat{\varphi}(\mathbf{x}, t) \hat{\varphi}(\mathbf{x}', t')) | 0 \rangle,$$

where the symbol T signifies the time ordered product of the operators,

$$T(\hat{\varphi}(\mathbf{x}, t) \hat{\varphi}(\mathbf{x}', t')) = \begin{cases} \hat{\varphi}(\mathbf{x}, t) \hat{\varphi}(\mathbf{x}', t') & \text{for } t > t' \\ \hat{\varphi}(\mathbf{x}', t') \hat{\varphi}(\mathbf{x}, t) & \text{for } t < t' \end{cases}$$

This expectation value is also called the **Feynman propagator**. One can visualize it in the above drawing by first considering complex values of t along the contour C_F , and then taking the limit when this contour is brought to the real axis; D_F corresponds to the values of D along C_F .

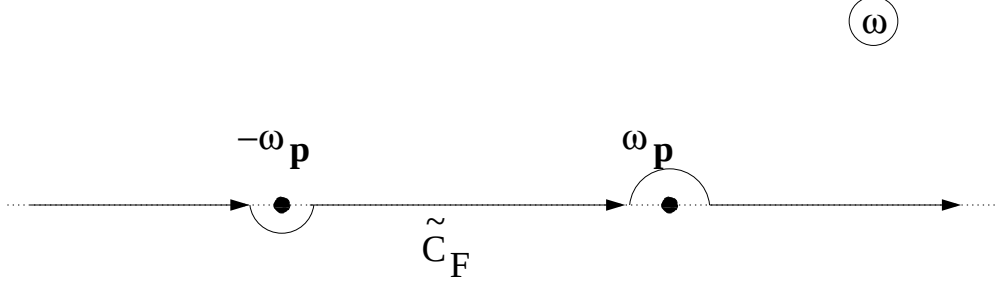
The Feynman propagator can be written as

$$D_F(\mathbf{x}, t) = \begin{cases} D_-(\mathbf{x}, t) = \int d\mu(\mathbf{p}) e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} & \text{for } t > 0 \\ D_+(\mathbf{x}, t) = \int d\mu(\mathbf{p}) e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}} & \text{for } t < 0 \end{cases},$$

but the most convenient representation is given by the 4-dimensional integral

$$D_F(\mathbf{x}, t) = \int_{\tilde{C}_F} \frac{d\omega}{2\pi} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{i e^{-i\omega t + i\mathbf{p} \cdot \mathbf{x}}}{\omega^2 - \mathbf{p}^2 - m^2},$$

The integrand has two poles, at $\omega = \pm\omega_{\mathbf{p}}$, and the integration contour $\tilde{C}_F$ for ω goes around these poles like this



If $t > 0$ the integrand decays exponentially into the lower half-plane of the variable ω , and one can perform the integration over this variable by closing the contour around $\omega = \omega_{\mathbf{p}}$. The integral reduces to the residue at this pole, which yields $D_{-}(\mathbf{x}, t)$. Similarly, when $t < 0$ the integration contour can be closed around the pole $\omega = -\omega_{\mathbf{p}}$, and the residue calculation yields $D_{+}(\mathbf{x}, t)$.

This integration prescription is usually expressed as

$$D_{\text{F}}(x) = \int \frac{d\omega}{2\pi} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{i e^{-i\omega t + i\mathbf{p}\mathbf{x}}}{\omega^2 - \mathbf{p}^2 - m^2 + i0} = \int \frac{d^4p}{(2\pi)^2} \frac{i e^{-ip_{\mu}x^{\mu}}}{p_{\mu}p^{\mu} - m^2 + i0}.$$

This form of the Feynman propagator is explicitly covariant. It also shows that $D_{\text{F}}(x)$ is the Green's function of the KG equation,

$$(\partial_{\mu}\partial^{\mu} + m^2) D_{\text{F}}(x) = -i \delta^{(4)}(x).$$

Note that the functions $D_{\pm}(x)$ satisfy homogeneous KG equation

$$(\partial_{\mu}\partial^{\mu} + m^2) D_{\pm}(x) = 0,$$

since in the KG theory the Heisenberg operator $\hat{\varphi}(x)$ itself satisfies homogeneous KG equation.

Exercise. The Feynman propagator is defined as

$$D_{\text{F}}(\mathbf{x}, t) = \theta(t) \langle 0 | \hat{\varphi}(\mathbf{x}, t) \hat{\varphi}(\mathbf{0}, 0) | 0 \rangle + \theta(-t) \langle 0 | \hat{\varphi}(\mathbf{0}, 0) \hat{\varphi}(\mathbf{x}, t) | 0 \rangle.$$

Applying the differential operator $\partial_{\mu}\partial^{\mu} + m^2$ and using the equal-time commutation relations

$$[\hat{\pi}(\mathbf{x}, t), \hat{\varphi}(\mathbf{x}', t)] = -i \delta^{(3)}(\mathbf{x} - \mathbf{x}'),$$

confirm that

$$(\partial_{\mu}\partial^{\mu} + m^2) D_{\text{F}}(x) = -i \delta^{(4)}(x).$$

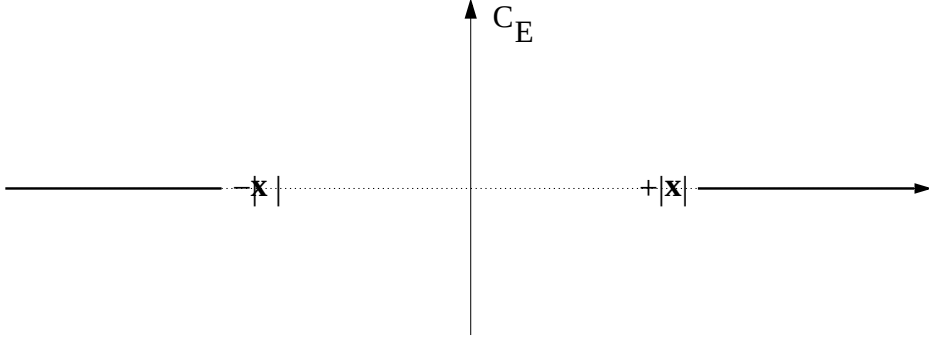
Finally, let us consider the “euclidean”, or “imaginary time” correlation function. It is defined in terms of the function $D(\mathbf{x}, t)$ taken at pure imaginary values of the time variable. Namely, let

$$t = -i x_4,$$

with real x_4 . Then

$$D_E(\mathbf{x}, x_4) = D(\mathbf{x}, -i x_4),$$

i.e. D_E is the function D with t taken along the contour C_E in the complex t -plane.



Note that this contour can be obtained from C_F by 90° rotation.

As is well known, by taking pure imaginary values of t one converts the Minkowski space-time into 4-dimensional Euclidean space, since

$$-d\tau^2 \rightarrow ds^2 = dx_4^2 + d\mathbf{x}^2,$$

and the Lorentz symmetry becomes the orthogonal group $O(4)$ of rotations of the the Euclidean 4-vectors

$$x_E = (\mathbf{x}, x_4)$$

The function $D_E(x_E)$ is called the Euclidean correlation function of the KG theory. We will discuss its significance later. For now, let us observe that since D_E is related to D_F simply by rotating the time contour, it satisfies the Euclidean-space version of the inhomogeneous KG equation

$$\left(m^2 - \frac{\partial^2}{\partial x_4^2} - \nabla^2\right) D_E(\mathbf{x}, x_4) = \delta(x_4) \delta^{(3)}(\mathbf{x}),$$

or in symmetric form

$$(m^2 - \Delta_E) D_E(x_E) = \delta^{(4)}(x_E),$$

where Δ_E stands for the 4-dimensional Laplacian. This equation is easily solved by Fourier transformation

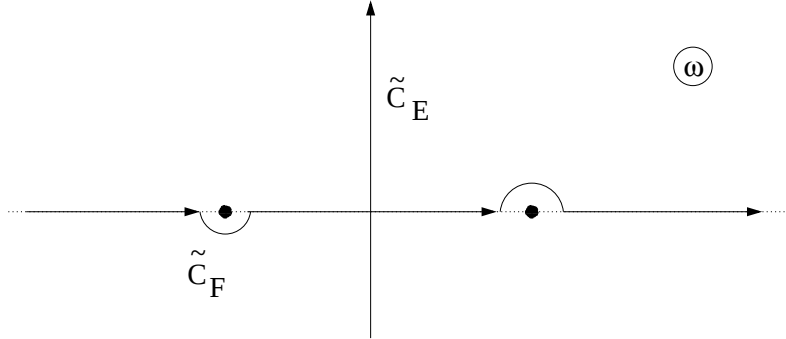
$$D_E(x_E) = \int \frac{d^4 p_E}{(2\pi)^4} \frac{e^{i p_E x_E}}{p_E^2 + m^2},$$

where the integration is over the 4-dimensional Euclidean momentum space, $p_E = (\mathbf{p}, p_4)$, and $p_E^2 = \mathbf{p}^2 + p_4^2$. Note that now the integrand does not have singularities in the integration domain. Also, unlike the Minkowski space version of this equation, the solution is unique if we demand that $D_E(x_E)$ decays at the Euclidean infinity $x_E \rightarrow \infty$. By introducing auxiliary integration, one can write this function as

$$D_E(x_E) = \int_0^\infty d\tau \int \frac{d^4 p_E}{(2\pi)^4} e^{-\tau(p_E^2 + m^2)} e^{i p_E x_E}, \quad (4.1)$$

This form is known as the Schwinger's proper-time representation.

Clearly, the above Euclidean momentum space integral for $D_E(x_E)$ is related to the covariant 4-momentum integral for the Feynman propagator. The correlation function $D_E(\mathbf{x}, x_4)$ is the analytic continuation of $D_F(\mathbf{x}, t)$ obtained by rotating the time contour C_F to C_E . Correspondingly, the integration contour $\tilde{C}_F$ in the complex ω -plane is rotated by 90° to the contour $\tilde{C}_E$.



This example suggests that some analytic characteristics of the KG theory, and perhaps more general QFT, may take simpler and more symmetric form when continued to pure imaginary time. This idea can be refined with the help of the path integral formulation of quantum mechanics, which we consider next.

5 Lecture 5

Let me briefly review the method of path integral in ordinary quantum mechanics. Here I will combine it with the idea of analytic continuation in the time variable. We will start with the "time evolution" in pure imaginary values of the time parameter. (Besides technical advantage of improving convergence of the integral, the imaginary-time path integral makes evident remarkable relation to classical statistical mechanics.) The real-time transition amplitudes can be recovered by analyticity. For derivation directly in real time see Sect.9.1 of PS.

Path integral in Quantum Mechanics

Given a quantum mechanical system, with the space of states $\mathcal{H}$ and the time-independent Hamiltonian $\hat{H}$, most problems can be reduced to the calculations of appropriate matrix elements of its time-evolution operator,

$$\langle f | e^{-\frac{i}{\hbar} \hat{H} t} | i \rangle .$$

In terms of the stationary states $| n \rangle$ this can be written as

$$\sum_n \langle f | n \rangle \langle n | i \rangle e^{-\frac{i}{\hbar} E_n t} .$$

In a typical problem, the energy spectrum E_n is bounded from below, $E_n \geq E_0$, and E_n grow to infinity with n . The above expression is a sum of oscillating terms, and we would like to have an efficient mathematical tool to sort them out. It is useful to consider this sum at complex values of t . If the number of states does not grow exponentially with E (and in ordinary quantum system it never does), this sum converges absolutely if t has negative imaginary part; thus it defines an analytic function of t in the lower half-plane. The real-time matrix element is the limiting value of this analytic function. One can start with pure imaginary t ,

$$t = -i \tau ,$$

and try to evaluate the matrix elements of the corresponding bounded Hermitian operator

$$\langle f | e^{-\frac{1}{\hbar} \hat{H} \tau} | i \rangle ,$$

anticipating that the real-time answer will be then obtained by analytic continuation.

Consider a system with one degree of freedom q , and assume that the Hamiltonian operator is “local” in the basis which diagonalizes $\hat{q}$; this means that the matrix elements

$$\langle q | \hat{H} | q' \rangle$$

have a support restricted to $q = q'$. Most typical Hamiltonian belong to this class, for instance for $H = \frac{1}{2} p^2 + V(q)$ the above matrix element is ($\hbar = 1$ from now on)

$$-\frac{1}{2} \delta''(q - q') + V(q) \delta(q - q').$$

In such cases it is possible to develop a path-integral representation for the matrix element

$$\langle q_f | e^{-\hat{H} \tau} | q_i \rangle.$$

The composition property

$$\langle q_f | e^{-\hat{H}(\tau_1 + \tau_2)} | q_i \rangle = \int \langle q_f | e^{-\hat{H} \tau_1} | q \rangle \langle q | e^{-\hat{H} \tau_2} | q_i \rangle dq$$

can be applied repeatedly to reduce the above matrix element to the composition of the matrix elements with arbitrarily small “time” intervals $\Delta\tau$,

$$\langle q_f | e^{-\hat{H} \tau} | q_i \rangle = \int \prod_{k=1}^{n-1} dq_k \prod_{k=1}^n \langle q_k | e^{-\hat{H} \Delta\tau} | q_{k-1} \rangle,$$

where $q_0 = q_i$, $q_n = q_f$, and $\Delta\tau = \tau/n$. In the limit of small $\Delta\tau$, as the consequence of the “locality” of $\hat{H}$ in the q -space, only the matrix elements $\langle q' | \exp(-\hat{H} \Delta\tau) | q \rangle$ with q' close to q will bring significant contributions to the integral.

To see this, consider again the Hamiltonian of the form

$$\hat{H} = \frac{1}{2} \hat{p}^2 + V(\hat{q}).$$

It is straightforward to show that in this case the matrix element

$$G(q, q' | \tau) = \langle q | e^{-\hat{H} \tau} | q' \rangle$$

satisfy the differential equation of the diffusion type

$$-\frac{\partial}{\partial \tau} G(q, q' | \tau) = \left(-\frac{1}{2} \frac{\partial^2}{\partial q^2} + V(q) \right) G(q, q' | \tau),$$

with the initial condition

$$G(q, q'|0) = \delta(q - q').$$

For $V(q) = 0$ the problem is easily solved,

$$G(q, q'|\tau) = \frac{1}{\sqrt{2\pi\tau}} e^{-\frac{(q-q')^2}{2\tau}}.$$

This function indeed decays fast at $|q - q'| \gg \sqrt{\tau}$. In the case of nonzero potential V the problem still can be examined in the limit of small τ . We write

$$G(q, q'|\tau) = e^{-\sigma(q, q'|\tau)}$$

and look for σ of the form of the following small- τ expansion

$$\sigma(q, q'|\tau) = \frac{\sigma_0(q, q')}{\tau} + a \log \tau + b + \tau \sigma_1(q, q') + \dots$$

Substituting this ansatz into the above diffusion equation one finds (I suggest doing it as an **Exercise**)

$$\begin{aligned} \sigma_0(q, q') &= \frac{(q - q')^2}{2}, & a &= 1/2, \\ \sigma_1(q, q') &= \frac{1}{q - q'} \int_{q'}^q V(y) dy, & \text{etc.} \end{aligned}$$

The leading terms do not depend on V , and are the same as for the free case, and σ_1 is the potential V averaged over the interval $[q', q]$. The constant $b = \frac{1}{2} \log 2\pi$ is determined by the initial condition at $\tau = 0$.

Because of the factor $\exp(-\sigma_0(q, q')/\tau)$ the function $G(q, q'|\tau)$ at small τ is sharply peaked around $|q - q'| = 0$ with the width $\sqrt{\tau}$. This is typical for the *Brownian motion*, where $\bar{R} \sim \sqrt{t}$ (our equation *is* the diffusion equation). If the potential V is differentiable (which we assume), we have

$$\sigma_1(q, q') = \frac{1}{2} V(q) + \frac{1}{2} V(q') + O(\tau)$$

(I have used the fact that $(q - q')^2 \sim \tau$). Therefore, for finite τ but large n (i.e. small $\Delta\tau = \tau/n$) we can write

$$\langle q_f | e^{-\hat{H}\tau} | q_i \rangle = \lim_{n \rightarrow \infty} \left(\frac{1}{2\pi \Delta\tau} \right)^{n/2} \int \left(\prod_{k=1}^{n-1} dq_k \right) e^{-\mathcal{A}\{q_k\}},$$

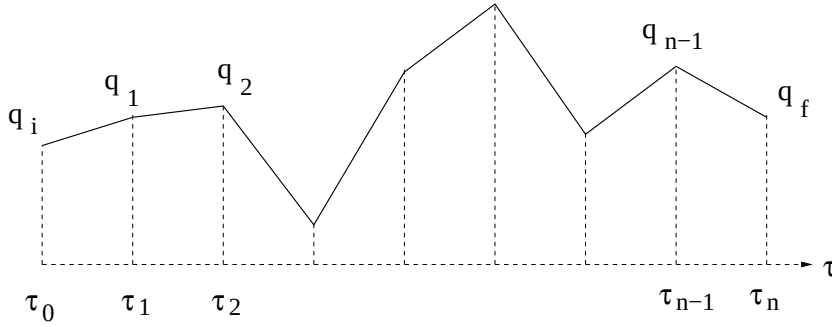
where

$$\mathcal{A}\{q_k\} = \sum_{k=1}^n \left[\frac{(q_k - q_{k-1})^2}{2 \Delta\tau} + \Delta\tau \frac{V(q_k) + V(q_{k-1})}{2} \right],$$

and $\Delta\tau = \tau/n$, and it is understood that

$$q_0 = q_i, \quad \text{and} \quad q_n = q_f.$$

Before the limit $n \rightarrow \infty$ is taken, the integration in this formula can be understood as going over the piecewise linear “trajectories” $q(\tau)$ running from q_i to q_f .



When $n \rightarrow \infty$, $\Delta\tau \rightarrow 0$, the function $\mathcal{A}\{q_k\}$ in the exponential has formal limit

$$\mathcal{A}[q(\tau)] = \int_0^\tau d\tau \left[\frac{1}{2} (dq(\tau)/d\tau)^2 + V(q(\tau)) \right].$$

It coincides with the “imaginary time action”, i.e. the action associated with our system, with t replaced by $-i\tau$,

$$e^{\frac{i}{\hbar} S[q(t)]} : \quad - \quad (t \rightarrow -i\tau) \quad \rightarrow \quad e^{-\mathcal{A}[q(\tau)]}.$$

It is conventional to write the $n \rightarrow \infty$ limit of the above n -fold integral as

$$\langle q_f | e^{-\hat{H}(\tau_f - \tau_i)} | q_i \rangle = \int [Dq(\tau)]_{\substack{q(\tau_i)=q_i \\ q(\tau_f)=q_f}} e^{-\mathcal{A}[q(\tau)]},$$

and to call it the (imaginary time) path integral, since the r.h.s involves the continuous path $q(\tau)$ from q_i to q_f .

Few remarks.

1. The path integral method makes the most explicit use of the *superposition principle* of quantum theory. The latter states that if some process can go several distinct ways, the full transition amplitude is the sum of the amplitudes associated with each possible way. Here (when returning back to real time) we sum the amplitudes associated with all distinct paths from q_i to q_f .

2. What kind of paths enter the path integral in the limit $n \rightarrow \infty$? Although it is correct to think of the limiting paths as continuous functions $q(\tau)$, absolute majority of the paths entering the integral are not smooth curves. This can be observed directly from the fact that for $\tau_1 - \tau_2 \rightarrow 0$ we have $|q(\tau_1) - q(\tau_2)| \sim \sqrt{|\tau_1 - \tau_2|}$, not $\sim |\tau_1 - \tau_2|$ as should be true for a continuously differentiable function. From this point of view the two terms in the action play significantly different role. The "kinetic" term $1/2 \int (dq/d\tau)^2 d\tau$ selects the class of the paths entering the integral, i.e. those paths for which $(q(\tau + \Delta\tau) - q(\tau))^2/\Delta\tau$ remains finite as $\Delta\tau \rightarrow 0$; in this sense the factor $\exp(-1/2 \int (dq/d\tau)^2 d\tau)$ should be considered as a part of the functional measure. The factor $\exp(-\int V(q(\tau)) d\tau)$ weights the paths according to their average potential energy.

3. Note that the symbol

$$[Dq(\tau)] = \lim_{n \rightarrow \infty} \left(\frac{1}{2\pi \Delta\tau} \right)^{n/2} \prod_{k=1}^{n-1} dq_k$$

contains divergent (in the limit) factor

$$\left(\frac{1}{2\pi \Delta\tau} \right)^{n/2} = e^{-\frac{\log(2\pi \Delta\tau)}{\Delta\tau} L},$$

where $L = (\tau_f - \tau_i) = n \Delta\tau$ is the "volume" of the imaginary time interval. This factor is reminiscent of the irritating infinity we had for the vacuum energy in the KG theory, and absorbing it into $[Dq(\tau)]$ is analogous to subtracting E_0 from $\hat{H}$.

4. We have derived the above path integral representation of the evolution operator in the imaginary time τ . But one can repeat the above calculation step by step, considering the paths in the real time t instead of τ (this is the form in which the path integral is usually introduced in quantum mechanics, see for instance §9.1 of the textbook). In this way one arrives at the expression

$$\langle q_f | e^{-\frac{i}{\hbar} H T} | q_i \rangle = \int [Dq(t)] e^{\frac{i}{\hbar} S[q(t)]},$$

with the phase $e^{\frac{i}{\hbar}S}$ replacing $e^{-\mathcal{A}}$. This of course is equivalent approach, although, unlike the imaginary time scheme, the integrals involved are not absolutely convergent. Usually, the most efficient way to handle such integrals is through analytic continuation (in variables or parameters) which makes the integral absolutely convergent, with subsequent taking the limiting values. The imaginary-time approach simply makes this procedure explicit.

5. Generalization of the above construction to the systems with many degrees of freedom q_a is straightforward, if the Hamiltonian is of the form

$$H = \sum_a \frac{1}{2} p_a^2 + V(q_a);$$

one obtains

$$\int [Dq_a] e^{-\mathcal{A}[q_a(\tau)]}$$

for the evolution operator. In more complicated cases the path integral in the phase space can be useful; this is briefly discussed in Sect.9.1 of PS.

Relation to Classical Statistical Mechanics

At this point, let me deviate to discuss remarkable relation which exists between the *Path Integral in Quantum Mechanics* (in its imaginary time version) and the *Classical Statistical Mechanics*. Let us recall some generalities of the classical statistical mechanics.

Suppose we have a classical dynamical system with the phase-space coordinates $\{P_i, Q_i\}$ (I will use capitals here because the relation between these Q_i and q_α appearing in the path integral is not going to be straightforward), and the Hamiltonian $\mathcal{H}(P_i, Q_i)$. In a thermal equilibrium state at a temperature T the probability distribution of microscopic states is given by the Gibbs formula

$$\mathcal{P}(P_i, Q_i) \prod_i dP_i dQ_i = Z^{-1} e^{-\beta \mathcal{H}(P_i, Q_i)} \frac{1}{N!} \prod_i dP_i dQ_i ,$$

where $\beta = 1/kT$, and Z is the partition function

$$Z = \int \frac{\prod_i dP_i dQ_i}{N!} e^{-\beta \mathcal{H}(P_i, Q_i)} .$$

Calculating the partition function is one of the main problems in statistical mechanics because the thermodynamic quantities are expressed through Z , say the free energy is $\beta F = -\log Z$. Again, a typical form of the Hamiltonian is

$$\mathcal{H}(P_i, Q_i) = \sum_i \frac{1}{2} P_i^2 + U(\{Q_i\}) .$$

In this case the integration over P_i is easy to perform,

$$Z = \frac{1}{N!} \left(\frac{2\pi}{\beta} \right)^{\frac{N}{2}} Z_{\text{conf}} ,$$

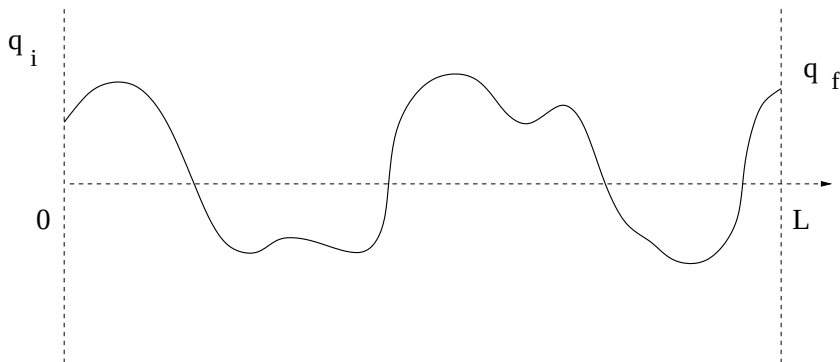
where N is the number of the degrees of freedom, $i = 1, 2, \dots, N$, and Z_{conf} is the configuration-space integral

$$Z_{\text{conf}} = \int \prod_i dQ_i e^{-\beta U(\{Q_i\})}$$

which contains most of interesting physics.

As a special example of such system, consider an elastic string whose configuration is described by a function $q(\tau)$, $\tau \in [0, L]$; the values $q(\tau)$ for all τ within this interval play the role of the configuration-space coordinates,

$$\{Q_i\} \rightarrow q(\tau) .$$



We also assume that the string lays in a potential well $V(q)$, so that the potential energy of the string is

$$U[q(\tau)] = \int_0^L d\tau \left[V(q(\tau)) + \frac{1}{2} \left(\frac{dq(\tau)}{d\tau} \right)^2 \right] ,$$

where the last term accounts for the elastic energy; to simplify the equations I have assumed that the elastic string tension equals 1.

To evaluate the configuration part of the partition function, we must integrate over the functions $q(\tau)$ satisfying given boundary conditions at the ends of the interval $[0, L]$, say

$$q(0) = q_i, \quad q(L) = q_f,$$

with the statistical weights $\exp(-\beta U[q(\tau)])$. Thus

$$Z_{\text{conf}} = \int [Dq(\tau)] e^{-\beta U[q(\tau)]}.$$

With the above form of $U[q(\tau)] = \mathcal{A}[q(\tau)]$, this expression is identical to the path integral for the imaginary time transition amplitude,

$$Z_{\text{conf}} \sim \langle q_f | e^{-\frac{1}{\hbar} \hat{H} L} | q_i \rangle,$$

provided we also identify

$$\beta = 1/\hbar.$$

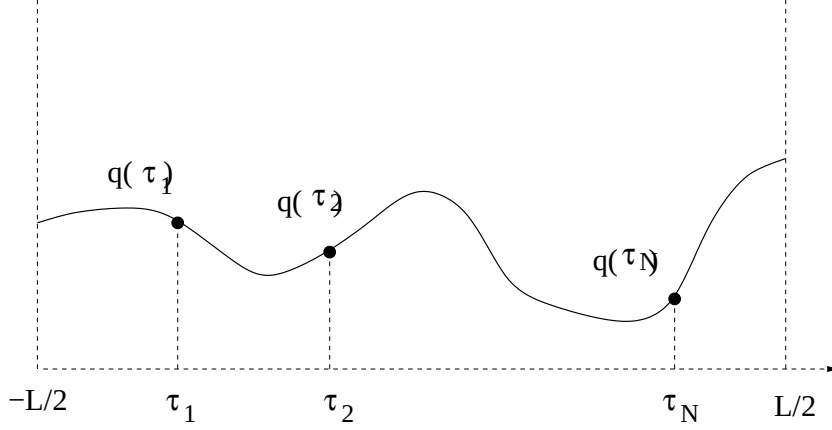
We see that quantum mechanics with a single degree of freedom q , and the Hamiltonian operator

$$\hat{H} = \frac{1}{2} \hat{p}^2 + V(\hat{q}),$$

is related in this way to classical statistical theory of a system with continuously many degrees of freedom $q(\tau)$.

Correlation functions

In statistical mechanics one may also be interested in *correlation functions*. In our example one may ask how the string positions $q(\tau)$ at different points, say at $\tau_1, \tau_2, \dots, \tau_N$, are correlated in the thermal equilibrium state. In the drawing below I made a trivial shift of the variable τ , so that from now on $\tau \in [-L/2, L/2]$.



The correlations are described by the averaged products $q(\tau_1)q(\tau_2)\dots q(\tau_N)$, which in turn are given by the integrals

$$\langle q(\tau_1) q(\tau_2) \dots q(\tau_N) \rangle = Z^{-1} \int q(\tau_1)q(\tau_2) \dots q(\tau_N) e^{-\mathcal{A}[q(\tau)]} [Dq(\tau)] ,$$

because $Z^{-1} e^{-\mathcal{A}[q(\tau)]} [Dq(\tau)]$ is the statistical weight of the microstate $q(\tau)$. It is instructive to find out how these correlation functions are interpreted in associated quantum mechanical problem.

Without loss of generality we can assume that

$$-L/2 \leq \tau_1 \leq \tau_2 \leq \dots \leq \tau_N \leq L/2 .$$

The integration over all microstates $q(\tau)$ can be performed in two steps: First, we fix the values of $q(\tau)$ at the points $\tau_1, \tau_2, \dots, \tau_N$ to be

$$q(\tau_1) = q_1 , \quad q(\tau_2) = q_2 , \quad \dots \quad q(\tau_N) = q_N ,$$

and integrate over all $q(\tau)$ subject to these constraints. After that, the integration over $q_1, q_1, \dots, q_N$ is performed to complete the integral. Comparing this with the path integral representation for the evolution operator, say

$$\langle q_1 \mid e^{-\hat{H}(\tau_2 - \tau_1)} \mid q_2 \rangle = \int [Dq(\tau)]_{\substack{q(\tau_1)=q_1 \\ q(\tau_2)=q_2}} e^{-\mathcal{A}[q(\tau)]}$$

we find

$$\langle q(\tau_1)q(\tau_2)\dots q(\tau_N) \rangle = Z^{-1} \int \left[\prod_{k=1}^N dq_k \right] \times q_1 q_2 \dots q_N \times$$

$$\langle q_f | e^{-\hat{H}(L/2-\tau_N)} | q_N \rangle \langle q_N | e^{-\hat{H}(\tau_N-\tau_{N-1})} | q_{N-1} \rangle \dots \langle q_1 | e^{-\hat{H}(\tau_1+L/2)} | q_i \rangle$$

Recall that in the coordinate representation the operator $\hat{q}$ acts by multiplication

$$\hat{q}\Psi(q) = q\Psi(q),$$

so that in this expression we can replace

$$\int dq | q \rangle q \langle q | = \hat{q}.$$

Then the above expression can be written as

$$\langle q_f | e^{-\hat{H}(L/2-\tau_N)} \hat{q} e^{-\hat{H}(\tau_N-\tau_{N-1})} \hat{q} \dots \hat{q} e^{-\hat{H}(\tau_1+L/2)} | q_i \rangle$$

which is best expressed in terms of the (imaginary time version of) Heisenberg operators

$$\hat{q}_E(\tau) = e^{\hat{H}\tau} \hat{q} e^{-\hat{H}\tau}$$

($\tau = it$), allowing one to write the above formula as

$$\langle q_f | e^{-\hat{H}\frac{L}{2}} \hat{q}_E(\tau_N) \hat{q}_E(\tau_{N-1}) \dots \hat{q}_E(\tau_1) e^{-\hat{H}\frac{L}{2}} | q_i \rangle$$

To simplify things further, let's take the limit $L \rightarrow \infty$, thus turning to the correlation functions of the infinite string. In this limit the exact boundary conditions, e.g. the values q_i and q_f , become unimportant. Indeed, the states $| q_i \rangle$ and $| q_f \rangle$ can be expanded in the stationary states $| n \rangle$ of the Hamiltonian $\hat{H}$, say

$$| q_i \rangle = \sum_n | n \rangle \langle n | q_i \rangle = \sum_n | n \rangle \Psi_n^*(q_i),$$

where it is assumed that

$$\langle n | m \rangle = \delta_{n,m}.$$

In the limit $L \rightarrow \infty$ the contribution of the ground state $| 0 \rangle$ dominates, since all other states are suppressed by the factor $\sim e^{-(E_n-E_0)\frac{L}{2}}$. Thus we have

$$\langle q(\tau_1)q(\tau_2)\dots q(\tau_N) \rangle = \frac{\langle q_f | 0 \rangle \langle 0 | q_i \rangle}{Z} e^{-E_0 L} \langle 0 | \hat{q}_E(\tau_N) \hat{q}_E(\tau_{N-1}) \dots \hat{q}_E(\tau_1) | 0 \rangle$$

Furthermore, note that

$$Z = \langle q_f | e^{-\hat{H}L} | q_i \rangle$$

hence the first factor in the above equation equals 1 (in the limit $L = \infty$), and we have

$$\langle q(\tau_1) \dots q(\tau_N) \rangle = \langle 0 | \hat{q}_E(\tau_N) \hat{q}_E(\tau_{N-1}) \dots \hat{q}_E(\tau_1) | 0 \rangle.$$

Note that the operators here are arranged in the order of τ increasing from the left to the right. Let me stress that the Euclidean Heisenberg operators can not be arranged any other way, as otherwise the product of the operators (the sum over the intermediate states) does not converge. More generally, if we consider complex values of τ_i (which we will), the above product of the operators makes direct sense in the domain

$$\Re \tau_1 \leq \Re \tau_2 \leq \dots \leq \Re \tau_N$$

where the sums over the intermediate states converge (hence the sums yield analytic function).

To relate the above Euclidean correlation function to the real-time ones one analytically continues to pure imaginary values of τ ,

$$\tau_k = it_k + 0,$$

where ”+0” shift indicates that if there is a branch cut along the imaginary axis the values at the right edge are to be taken.

6 Lecture 6

Previously, we have discussed the path integral representation in ordinary quantum mechanics. The real-time transition amplitude is represented as a suitably defined integral

$$\langle q_f | e^{-\frac{i}{\hbar} H T} | q_i \rangle = \int [Dq(t)]_{\substack{q(0)=q_i \\ q(T)=q_f}} e^{\frac{i}{\hbar} S[q(t)]},$$

over all "paths" $q(t)$ from q_i to q_f . Now I want to discuss how the idea of the path integral can be applied to a relativistic particle.

Path integral for a relativistic particle

We have seen that the KG theory is the theory of spinless Bose particles, and we expect that the Feynman propagator $D_F(x_i - x_f)$ (and related Euclidean correlation function D_E) admit an interpretation in terms of amplitude associated with the particle propagation from x_i to x_f . Can we write such amplitude as an integral over the "paths"?

In classical mechanics the action of a relativistic particle travelling from x_i to x_f is

$$S = -m \int_{x_i}^{x_f} \sqrt{dx_\mu dx^\mu} = -m \int_{(t_i, \mathbf{x}_i)}^{(t_f, \mathbf{x}_f)} \sqrt{dt^2 - d\mathbf{x}^2},$$

where the integration is performed along a given path (generally described by a parametric curve $x^\mu(s)$) from x_i to x_f . In classical theory it is assumed that $(dx^0)^2 \geq d\mathbf{x}^2$ along the whole path (the particle velocity never exceeds the speed of light) the time component $t = x^0$ of x^μ can serve as the parameter of the path, and we can write

$$S[\mathbf{x}(t)] = -m \int_{t_i}^{t_f} dt \sqrt{1 - \dot{\mathbf{x}}^2}.$$

In quantum theory we may attempt to write for the transition amplitude

$$\int_{\text{paths}(i \rightarrow f)} e^{\frac{i}{\hbar} S} [D\mathbf{x}(t)],$$

where the integration is over "all paths" from x_i to x_f .